

Complex Manifolds

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MA726

Abstract

MA722: Differential Geometry 2 was taught by my [Professor Matiej Szczesny](#) in the spring 2026 semester (my junior year). Although not officially a topics course, it is treated by the department as a topics course and the officially listed course description is not accurate. This semester's topic is complex geometry, and we followed the book [\[Huy04\]](#).

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Chapter 1

Local theory

1.1 Holomorphic functions of several variables

Lecture 1: Syllabus day

Tue 20 Jan 2026 12:25

Book: *Complex Geometry* by Daniel Hugbrechts.

Definition 1.1.1. A complex manifold is a topological manifold locally modeled by open subsets of $\mathbb{C}^n$ such that the transition functions are biholomorphic (holomorphic and invertible).

Holomorphic functions are extremely rigid. For example, they are necessarily analytic (stronger than C^∞ – has convergent power series). These are very special manifolds and we should be able to say a lot about them. However, we cannot, for example, have partitions of unity or compactly supported holomorphic functions, so the study of them is very different from that of real manifolds, and we will use many tools that show up in algebraic geometry.

Recall: let $f : U \subseteq \mathbb{C} \rightarrow \mathbb{C}$. Then f is holomorphic if any of the following equivalent characterizations hold.

1. f satisfies the Cauchy-Riemann (C-R) equations: Let $z = x + iy$ and $f(z) = u(x, y) + iv(x, y)$. Then $\partial_x u = \partial_y v$ and $\partial_y u = -\partial_x v$.
2. $dfJ = Jdf$ for $J \in \text{Hom}_{\mathbb{C}}(\mathbb{C}, \mathbb{C})$ the complex structure $x \mapsto ix$ (easy verification, multiply matrices in both directions and recover C-R eqns).
3. Another way of saying (2) is df is $\mathbb{C}$ -linear.
4. Another formulation is defining $\partial_z = \frac{1}{2}(\partial_x - i\partial_y)$ and $\partial_{\bar{z}} = \frac{1}{2}(\partial_x + i\partial_y)$. Then we require $\partial_{\bar{z}}f = 0$. This follows by expanding it out in terms of u, v, x, y , and recovering C-R.
5. Near all $c \in U$, there exists an ε -ball $B(\varepsilon, c)$ centered at c where f has a convergent power series expansion.

We also recall some relevant theorems from single variable complex analysis.

Proposition 1.1.2 (Maximum principle). *Let $U \subseteq \mathbb{C}$ be open and connected. If $f : U \rightarrow \mathbb{C}$ is holomorphic and non-constant, then $|f|$ has no local maximum in U . If U is bounded and f can be extended to a continuous function $f : \bar{U} \rightarrow \mathbb{C}$, then $|f|$ takes its maximal values on the boundary ∂U .*

Proposition 1.1.3 (Identity theorem). *If $f, g : U \rightarrow \mathbb{C}$ are holomorphic on a connected open subset $U \subseteq \mathbb{C}$ such that for a non-empty open subset $V \subseteq U$, the restrictions $f|_V = g|_V$ agree, then $f = g$.*

Proposition 1.1.4 (Riemann extension theorem). *Let $f : B_\varepsilon(z_0) \setminus \{z_0\} \rightarrow \mathbb{C}$ be a bounded holomorphic function. Then f can be extended to a holomorphic function $f : B_\varepsilon(0) \rightarrow \mathbb{C}$.*

Proposition 1.1.5 (Riemann mapping theorem). *Let $U \subseteq \mathbb{C}$ be a simply connected proper open subset. Then U is biholomorphic to the unit ball $B_1(0)$, i.e. there exists a bijective holomorphic map $f : U \rightarrow B_1(0)$ such that its inverse is also holomorphic.*

Proposition 1.1.6 (Liouville). *Every bounded holomorphic function $f : \mathbb{C} \rightarrow \mathbb{C}$ is constant. In particular, there is no biholomorphic map between $\mathbb{C}$ and a ball $B_\varepsilon(0)$ with $\varepsilon < \infty$. This is a striking difference to the real situation and will cause a different concept of locality for complex manifolds than the one we are used to from real differential geometry.*

Proposition 1.1.7 (Residue theorem). *Let $f : B_\varepsilon(0) \setminus \{0\} \rightarrow \mathbb{C}$ be a holomorphic function. Then f can be expanded in a Laurent series*

$$f(z) = \sum_{n=-\infty}^{\infty} a_n z^n$$

and the coefficient a_{-1} is given by the residue formula

$$a_{-1} = \frac{1}{2\pi i} \int_{|z|=\varepsilon/2} f(z) dz.$$

The residue theorem is usually applied to more general situations where the function f has several isolated singularities in a connected open subset and the integral is taken over a closed contractible path surrounding the singularities.

We generalize this to several complex variables. We will not get into why this makes sense beyond contrasting with our previous characterizations since this subject is quite different than complex analysis in 1 variable.

We have notation as follows. $z \in \mathbb{C}^n$ is z_j with each component written as $z_j = x_j + iy_j$. We will often write i instead of j and assume it does not get confused with the complex unit. We will sometimes use $z^J = z_i^{j_k}$. A basis for the topology is given by polydisks

$$\Delta(\varepsilon, c) = \{z \in \mathbb{C}^n \mid |z_j - c_j| < \varepsilon_j \forall j = 1, \dots, n\}.$$

Polydisks in $\mathbb{C}^n$ are like real $2n$ -dimensional rectangles.

Theorem 1.1.8. *Let $U \subseteq \mathbb{C}^n$ open and $f \in C^\infty(U, \mathbb{C})$ (although assuming C^1 suffices). The following are equivalent.*

1. *f satisfies the C-R equations in each z_i : $\partial_{x_i} u = \partial_{y_i} v$ and $\partial_{x_i} v = -\partial_{y_i} u$;*
2. *f is separably holomorphic in each variable z_j , i.e. fixing all but one variable is holomorphic;*
3. *Given $c \in U$, there exists an ε -polydisk $\Delta(\varepsilon, c)$ such that $f(z)$ has a powerseries expansion in the polydisk $\Delta(\varepsilon, c)$*

$$f(z) = \sum_{|J|=0}^{\infty} a_J (z - c)^J.$$

Sketch of proof. Proof of (1) $\iff$ (2) is simple via 1-variable complex analysis. Power series converge on their interior, so they converge in any variable, thus (3) $\implies$ (2). The only work required to prove this is (2) $\implies$ (3). This follows from the multivariate Cauchy Integral Formula (CIF). $\blacksquare$

Definition 1.1.9. A function $f : U \subseteq \mathbb{C}^n \rightarrow \mathbb{C}$ is *holomorphic* if it satisfies any of the criteria of theorem 1.1.8.

Theorem 1.1.10 (Cauchy integration formula, [Huy04] 1.1.2). *Let f be holomorphic on a polydisk $D = \prod_j D_j$ with D_j an open disk, and continuous on its closure. Then for $z \in D$,*

$$f(z) = \frac{1}{(2\pi i)^n} \int \cdots \int_{\partial D_1 \times \cdots \times \partial D_n} \frac{f(\zeta_1, \dots, \zeta_n)}{\prod_j (\zeta_j - z_j)} d\zeta_1 \cdots d\zeta_n.$$

Many 1-variable theorems generalize. For example, the maximum modulus principle and the identity theorem. If f is a nonconstant holomorphic function, then its modulus does not attain a maximum on the interior of its domain. If f and g are analytic on a domain D and $f = g$ on some $S \subseteq D$ with an accumulation point, then $f = g$ on D .

Lemma 1.1.11 ([Huy04] 1.1.3). *Let $U \subseteq \mathbb{C}^n$ be open, and let $V \subseteq \mathbb{C}$ be an open neighborhood of the boundary of an ε -sphere $\partial B_\varepsilon(0)$. Suppose $f : U \times V \rightarrow \mathbb{C}$ is holomorphic. Then the function*

$$g(z_1, \dots, z_n) = \int_{|\zeta|=\varepsilon} f(\zeta, z_1, \dots, z_n) d\zeta$$

is holomorphic on U .

Sketch of proof. Differentiate under the integral sign; this is valid by compactness $|\zeta| = \varepsilon$, or alternatively the dominated convergence theorem. Thus

$$\frac{\partial g}{\partial \bar{z}_i} = \int_{|\zeta|=\varepsilon} \frac{\partial f}{\partial \bar{z}_i}(\zeta, z_1, \dots, z_n) d\zeta.$$

Since f is holomorphic, appealing to theorem 1.1.8 (2) the interior of the integral vanishes, and g is holomorphic also by theorem 1.1.8 (2). $\blacksquare$

More review.

Definition 1.1.12. Let $(X, \mathcal{T})$ be a topological space and $\text{Open}(X)$ the poset category $\mathcal{T}$. Then a *presheaf* with values in a *concrete* category $\mathcal{C}$ is a functor $\mathcal{F} : \text{Open}(X)^{\text{op}} \rightarrow \mathcal{C}$.

Often we take $\mathcal{C} = \text{Set}, \text{Ab}, R\text{-Mod}$, or another concrete abelian category $\mathcal{A}$. If $U \subseteq X$ is open, we call elements of $\mathcal{F}(U)$ *sections* of $\mathcal{F}$ over U . If $U \subseteq V$ are open in X and $s \in \mathcal{F}(V)$, then we write $s|_U$ for the image of s in the induced map $\mathcal{F}(V) \rightarrow \mathcal{F}(U)$.

Definition 1.1.13. A *sheaf* is a presheaf $\mathcal{F}$ that satisfies the following axioms.

- (Locality) If $U \subseteq X$ is open and $\{U_i \subseteq U\}_{i \in I}$ is an open cover of U , and $s, t \in \mathcal{F}(U)$ have $s|_{U_i} = t|_{U_i}$ for all i , then $s = t$.
- (Gluing) Let $U \subseteq X$ be open, $\{U_i \subseteq U\}_{i \in I}$ an open cover of U and $\{s_i \in \mathcal{F}(U_i)\}_{i \in I}$ a collection of sections. If $s_i|_{U_i \cap U_j} = s_j|_{U_i \cap U_j}$ for all $i, j \in I$, then there is some $s \in \mathcal{F}(U)$ such that $s_i = s|_{U_i}$ for all i .

Locality and gluing can be reinterpreted in terms of exactness of a sequence, and even more generally this can be rephrased as a certain diagram being an equalizer diagram. We will not need non-concrete categories in this case, so we ignore this generalization.

Notation 1.1.14. Given a space X , we define the *category of $\mathcal{C}$ -valued presheaves* as $\mathcal{Pshf}(X, \mathcal{C}) := \mathcal{C}^{\mathcal{O}pen(X)^{op}}$. We define the category $\mathcal{Shf}(X, \mathcal{C})$ of $\mathcal{C}$ -valued sheaves on X as the full subcategory of $\mathcal{Pshf}(X, \mathcal{C})$ spanned by sheaves. This means that a morphism of (pre)sheaves are natural transformations.

Definition 1.1.15. Let $\mathcal{F}$ be a (pre)sheaf of a space X on a category $\mathcal{C}$. The *stalk* $\mathcal{F}_p$ of $\mathcal{F}$ at a point $p \in X$ is the quotient

$$\{(s, U) \mid U \in \mathcal{O}pen(X), p \in U, s \in \mathcal{F}(U)\} / \sim$$

where $\sim$ is defined by $(s, U) \sim (t, V)$ iff there is an open set $W \subseteq U, V$ with $p \in W$ such that $s|_W = t|_W$. A *germ* of $\mathcal{F}$ at p is an element of the stalk of $\mathcal{F}$ at p . An equivalent characterization of the stalk is as follows. Let $\mathcal{U}_p \subseteq \mathcal{O}pen(X)$ be the full subcategory spanned by open sets containing p . Then

$$\mathcal{F}_p := \text{colim } \mathcal{F}|_{\mathcal{U}_p}.$$

Proposition 1.1.16. *Let $\mathcal{A}$ be an abelian category and X a space. Then $\mathcal{Pshf}(X, \mathcal{A})$ is abelian.*

Proof. By [Mac98], limits in functor categories exist and are computed pointwise. All statements that remain to be shown are very straightforward. ■

Corollary 1.1.17. *Let $\mathcal{A}$ be an abelian category and X a space. Then $\mathcal{Shf}(X, \mathcal{A})$ is abelian.*

Proof. It suffices to show sheaves are closed under taking (co)limits, which is trivial. ■

Example 1.1.18 ([Huy04] 1.1.14). Let $\mathcal{O}_{\mathbb{C}^n}$ denote the sheaf of holomorphic functions, where the sections of $\mathcal{O}_{\mathbb{C}^n}$ over $U \subseteq \mathbb{C}^n$ are all holomorphic functions $f : U \rightarrow \mathbb{C}$, and for each $U \subseteq V$ the restriction morphism $\mathcal{O}_{\mathbb{C}^n}(V) \rightarrow \mathcal{O}_{\mathbb{C}^n}(U)$ is precisely function restriction, meaning we map $f \in \mathcal{O}_{\mathbb{C}^n}(V)$ as $f \mapsto f|_U$. We write the stalk of $\mathcal{O}_{\mathbb{C}^n}$ at $z \in \mathbb{C}^n$ as $\mathcal{O}_{\mathbb{C}^n, z}$.

Lecture 2: Multivariate holomorphic functions and Weierstrass polynomials

Thu 22 Jan 2026 12:24

Theorem 1.1.19 (Hartog, [Huy04] 1.1.4). *Let there exist nested polydisks $\Delta(\varepsilon') \subseteq \Delta(\varepsilon)$ in $\mathbb{C}^n$ for $n \geq 2$, which we can WLOG take to be centered at 0. Let $U := \Delta(\varepsilon) \setminus \Delta(\varepsilon')$ and let $f : U \rightarrow \mathbb{C}$ be holomorphic. Then f extends to $\Delta(\varepsilon)$.*

Proof. Recall the following for 1-variable functions: if $f(z)$ is analytic in an annulus centered at 0, then $f(z)$ can be written

$$f(z) = \sum_{n \in \mathbb{Z}} a_n z^n, \quad a_n = \frac{1}{2\pi i} \oint_C f(\xi) \xi^{-n-1} d\xi$$

for any contour C with winding number 1 around the annulus (aka anything in the homotopy class; there are no singularities in the annulus by holomorphy).

We prove it in 2 variables. With assumptions and notation as stated in the problem, we can do a Laurant expansion in each z_i with the rest fixed. Hence

$$f(z) = \sum_{n \in \mathbb{Z}} a_n(z_1) z_1^n, \quad a_n(z_1) = \frac{1}{2\pi i} \oint_C f(z_1, \xi) \xi^{-n-1} d\xi.$$

Applying lemma 1.1.11 allows us to realize this as a holomorphic function in z_2 in the entire disk, hence $a_n(z_1) = 0$ for $n < 0$ and $\varepsilon'_1 < |z_1| < \varepsilon_1$. By the 1-variable identity principle, this implies $a_n(z_1) = 0$ for $n < 0$ in the entire disk. Thus only non-negative powers of z_2 occur in the expansion. The same logic applies for z_1 . Hence f extends via power series to the entire polydisk, and is particular holomorphic on $\Delta(\varepsilon)$. It is clear how an inductive proof can now proceed, or how this logic can prove $n > 1$ dimensions automatically. ■

Remark 1.1.20. This implies that in 2 or more variables, singularities are never isolated, nor are bounded. They look more like long codimension 1 hypersurfaces, in particular branched covers of $\mathbb{C}^{n-1}$.

Remark 1.1.21. This is an analytic analogue of the fact from AG that $\mathbb{A}^n \setminus \{0\}$, $n \geq 2$ is not affine.

We now explore precisely what singularities of multivariate complex variables look like. By passing to the reciprocal we can turn singularities into zeroes, so we start by trying to understand zero sets of analytic functions in $\mathbb{C}^n$.

Definition 1.1.22. A *Weierstrass polynomial* in z_1 is a polynomial of the form

$$z_1^d + \alpha_1(z_2, \dots, z_n) z_1^{d-1} + \alpha_2(z_2, \dots, z_n) z_1^{d-2} + \dots + \alpha_d(z_2, \dots, z_n)$$

where each α_i is a holomorphic function in some small disk in $\mathbb{C}^{n-1}$ about the origin, and they vanish at $0 \in \mathbb{C}^{n-1}$.

Proposition 1.1.23 (Weierstrass perparation, [Huy04] 1.1.6). *Let $f : \Delta(\varepsilon, 0) \rightarrow \mathbb{C}$ be holomorphic with $f(0) = 0$ and $f(z_1, 0, \dots, 0)$ not identically zero. Then in a polydisk $\Delta(\varepsilon', 0)$ with $\varepsilon' < \varepsilon$ we can write f as $p \cdot u$, where p is a Weierstrass polynomial and $u(0) \neq 0$ is a unit in the ring of germs at 0: $u \in \mathcal{O}_{\mathbb{C}^n, 0}$. In particular, p contains all the information of the zero set of f .*

I omit the proof in exchange for intuition for the zero set of f . Take f a function on $\mathbb{C}^2$. Fix a variable z_2 ; the Weierstrass polynomial is now a polynomial of degree $d - 1 = 3$ in z_1 which has $d - 1 = 3$ roots in the relevant slice of $\mathbb{C}^2$. Now letting z_2 vary shows that there are $d - 1 = 3$ roots in each slice which vary smoothly, yielding 3 lines (counting multiplicity) of singularities.

Theorem 1.1.24 (Weierstrass division). *Under the assumptions of proposition 1.1.23, let $p(z_1, \dots, z_n)$ be a Weierstrass polynomial in z_1 of degree d . In a sufficiently small $\Delta(\varepsilon, 0)$ we can write $f = q \cdot p + s$, where q is holomorphic in $\Delta(\varepsilon, 0)$ and $s \in \mathcal{O}_{\mathbb{C}^{n-1}}[z_1]$ is a polynomial in z_1 with coefficients in $\mathcal{O}_{\mathbb{C}^{n-1}}$ evaluated at a polydisk sufficiently close to 0, and is of degree less than d in z_1 . Note that s need not be a Weierstrass polynomial.*

Now we consider algebraic properties of the ring of germs of $\mathcal{O}_{\mathbb{C}^n, 0}$. This can be identified with the ring $\mathbb{C}\{z_1, \dots, z_n\}$ of convergent power series in a sufficiently small polydisk in all n variables by passing to the power series representation of holomorphic functions.

Remark 1.1.25. This is a power series because $\mathbb{C}\{z_1, \dots, z_n\}$ is obviously an integral domain.

Remark 1.1.26. $\mathcal{O}_{\mathbb{C}^n,0}$ is a *local ring*, meaning there is a unique maximal ideal. This is given by functions that vanish at the origin. To check this, we need only show that it is a unit. This indeed follows since it must have a multiplicative inverse.

Proposition 1.1.27. $\mathcal{O}_{\mathbb{C}^n,0}$ is a UFD.

Proof. We induct on n . Take $n = 1$. The only irreducible element up to units is $f(z) = z$, and all holomorphic functions are analytic. Assume the inductive hypothesis for $n - 1$. Recall Gauss' lemma implies that if R is a UFD, then $R[x]$ is a UFD. Take $f \in \mathcal{O}_{\mathbb{C}^n,0}$ a non-unit, meaning $f(0) = 0$. By proposition 1.1.23, $f = p \cdot u$ for u a unit in $\mathcal{O}_{\mathbb{C}^n,0}$ and p a Weierstrass polynomial, which lives in $p \in \mathcal{O}_{\mathbb{C}^{n-1},0}[z_1]$. By induction and Gauss' lemma, $\mathcal{O}_{\mathbb{C}^{n-1},0}[z_1]$ is a UFD, so factor p as irreducible polynomials $p = p_1 \cdots p_r$. We claim p_i are all Weierstrass. I omit the proof. We now need to show the p_i s are irreducible in $\mathcal{O}_{\mathbb{C}^n,0}$, since we are working in a proper subring $\mathcal{O}_{\mathbb{C}^{n-1},0} \subsetneq \mathcal{O}_{\mathbb{C}^n,0}$.

Suppose for contradiction that $p_j = a \cdot b$ in $\mathcal{O}_{\mathbb{C}^n,0}$, meaning one of these must be a unit. By proposition 1.1.23 on each of a, b , we obtain $a = q_1 u_q$ and $b = q_2 u_2$. Then $p_j = q_1 u_1 q_2 u_2$. We claim $q_1 q_2$ is also a Weierstrass polynomial. We then appeal to the uniqueness part of proposition 1.1.23 which we did not state but it says the polynomial is unique. Hence $u_1 u_2 = 1$, and thus p_j is a product of Weierstrass polynomials. Thus contradicts irreducibility of p_j . ■

Lecture 3: a

Tue 27 Jan 2026 12:29

Office hours are Wednesday 1-3pm, cds431. MA822 will probably move to psy b33.

Proposition 1.1.28. $\mathcal{O}_{\mathbb{C}^n,0}$ is Noetherian.

Proof. We induct on n . At $n = 1$ the statement is trivial apparently. Assume the inductive hypothesis for $n - 1$. Recall that the Hilbert basis theorem says that N Noetherian implies $N[x]$ Noetherian. We apply this here to obtain $\mathcal{O}_{\mathbb{C}^{n-1},0}[z_1]$ is Noetherian. Let $I \subseteq \mathcal{O}_{\mathbb{C}^n,0}$ be a nonzero proper ideal. Then it contains some nonzero nonunit $f \in I$, so $f(0) = 0$. By proposition 1.1.23, $f = p \cdot u$ for p a Weierstrass polynomial and u a unit. Then $f \in I$ implies $f \cdot u^{-1} = p \in I$. Consider $I' := I \cap \mathcal{O}_{\mathbb{C}^{n-1},0}$. Then since $\mathcal{O}_{\mathbb{C}^{n-1},0}$ is Noetherian, $I' = \langle q_1, \dots, q_r \rangle$. We claim $I = \langle p, q_1, \dots, q_r \rangle$. Choose $g \in I$. Then theorem 1.1.24 divides g as $ap + s$ for $s \in \mathcal{O}_{\mathbb{C}^{n-1},0}$ of degree less than p and a holomorphic. Since $g, p \in I$, we have $s \in I$. Thus $s \in I'$, and thus s is a linear combination in q_i . Thus g is a linear combination in p and the q_i s, concluding the proof. ■

Remark 1.1.29. We do not need to only consider rings of germs at the origin, since up to translation any point can be mapped to the origin.

Definition 1.1.30. Let $U \subseteq \mathbb{C}^n$ be open. An *analytic subvariety* of U is a subset $V \subseteq U$ such that for every $z_0 \in V$, there exists a collection of holomorphic functions $\{f_i : U \rightarrow \mathbb{C}\}_{1 \leq i \leq r}$ such that z_0 is in the intersection of the vanishing loci of the f_i s.

We can then discuss germs of analytic varieties, and obtain a nice language similar to that of algebraic geometry for $\mathbb{C}^n$. We will not continue down this path, and will instead begin our discussion of complex geometry.

Definition 1.1.31. Let $U \subseteq \mathbb{C}^n$ and $V \subseteq \mathbb{C}^m$ open. Then $f : U \rightarrow V$ is holomorphic iff all f_j s are holomorphic. We say f is biholomorphic if it has a holomorphic inverse.

Remark 1.1.32. With notation as in definition 1.1.31, if f is a biholomorphism, then $n = m$ is forced. We do not prove this since it is not quite trivial.

Notation 1.1.33. We will generally write coordinates in the domain as $z_i = x_i + iy_i$ and coordinates in the codomain as $w_i = u_i + iv_i$.

Identifying $\mathbb{C}^n \cong \mathbb{R}^{2n}$, the derivative takes the form

$$Df = \begin{bmatrix} \frac{\partial u^i}{\partial x_j} & \frac{\partial u^i}{\partial y_j} \\ \frac{\partial v^i}{\partial x_j} & \frac{\partial v^i}{\partial y_j} \end{bmatrix}.$$

This is $\mathbb{C}$ -linear $T_a\mathbb{C}^n \rightarrow T_{f(a)}\mathbb{C}^m$, which are complex vector spaces now defined as $C^\infty(\mathbb{C}^n)$. In this case, Df is represented by the frames ∂_{x_i} and ∂_{y_i} by

$$Df_i^j = \frac{\partial f^j}{\partial z_i}.$$

For example, take $U, V \subseteq \mathbb{C}$. The fact that f is holomorphic and Df viewed as $\mathbb{R}^2 \rightarrow \mathbb{R}^2$ is $\mathbb{C}$ -linear means that for J the complex structure

$$J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix},$$

we have $J \circ Df = Df \circ J$. Expanding this out yields

$$Df(\partial_x) = \frac{\partial f}{\partial z} \partial_u.$$

It is convenient to work with complex tangent vectors.

Warning 1.1.34. The vectors ∂_{z_j} and $\partial_{\bar{z}_j}$ *do not live* in $T_a\mathbb{R}^{2n}$. This is because $\partial_{z_j} = \frac{1}{2}(\partial_{x_j} + i\partial_{y_j})$, and the number i doesn't make sense in $T_a\mathbb{R}^{2n}$. Thus we need to tensor with $\mathbb{C}$ over $\mathbb{R}$. Thus we usually consider $T_a\mathbb{C}^n \otimes_{\mathbb{R}} \mathbb{C}$.

Claim 1.1.35. *With respect to the frames ∂_{z_j} and $\partial_{\bar{z}_k}$, $Df : T_a\mathbb{C}^n \rightarrow T_{f(a)}\mathbb{C}^m$ extends to a linear transformation*

$$Df : T_a\mathbb{C}^n \otimes_{\mathbb{R}} \mathbb{C} \rightarrow T_a\mathbb{C}^n \otimes_{\mathbb{R}} \mathbb{C}.$$

Moreover it is of the form

$$Df = \begin{bmatrix} (\partial_z f) & (\partial_{\bar{z}} f) \\ (\partial_z \bar{f}) & (\partial_{\bar{z}} \bar{f}) \end{bmatrix}.$$

Thus if it is holomorphic,

$$Df = \begin{bmatrix} (\partial_z f) & 0 \\ 0 & (\partial_{\bar{z}} \bar{f}) \end{bmatrix} = \begin{bmatrix} (\partial_z f) & 0 \\ 0 & \overline{(\partial_z f)} \end{bmatrix}.$$

Let's see this when $m = n = 1$. Then

$$Df(\partial_z) = \begin{pmatrix} \partial_x u & \partial_y u \\ \partial_x v & \partial_y v \end{pmatrix} \begin{pmatrix} \frac{1}{2} \\ \frac{i}{2} \end{pmatrix} = \frac{1}{2} \begin{pmatrix} \partial_x u \\ \partial_x v \end{pmatrix} - \frac{i}{2} \begin{pmatrix} \partial_y u \\ \partial_y v \end{pmatrix} = \frac{1}{2} \begin{pmatrix} \partial_x u - i\partial_y u \\ \partial_x v - i\partial_y v \end{pmatrix}.$$

We claim this is

$$\frac{\partial f}{\partial z} \frac{\partial}{\partial w} + \frac{\partial \bar{f}}{\partial z} \frac{\partial}{\partial \bar{w}}.$$

This can be verified by brute force. We remark that since $T_a\mathbb{C}^n \otimes_{\mathbb{R}} \mathbb{C}$ is basically just $T_a\mathbb{C}^n \oplus iT_a\mathbb{C}^n$ and since the Jacobians of holomorphic functions are half zero, they basically reduce to maps $T_a\mathbb{C}^n \rightarrow T_a\mathbb{C}^n$ in the way we would conventionally think, and the determinant agrees with their real Jacobian.

Lecture 4: sldfkskm

Thu 29 Jan 2026 12:28

Theorem 1.1.36 (Implicit function, real version). *If $f : U \subseteq \mathbb{R}^n \rightarrow V \subseteq \mathbb{R}^m$, U, V open, and $Df_a : T_a \mathbb{R}^n \rightarrow T_{f(a)} \mathbb{R}^m$ is surjective for all $a \in f^{-1}(f(a))$, then if $f^{-1}(f(a))$ is the graph of a smooth function, then $f^{-1}(f(a))$ is a manifold.*

Theorem 1.1.37 (Lazarfeld 1.19; Inverse and implicit function). *Let $f : U \subseteq \mathbb{C}^n \rightarrow V \subseteq \mathbb{C}^m$ be holomorphic with a Jacobian that is surjective (at a iff neighborhood of a).*

1. *If $n = m$, then there exist neighborhoods $U' \ni a$ and $V' \ni f(a)$ such that $f|_{U'}$ is a biholomorphism to V' .*
2. *If $n > m$ then $f^{-1}(f(a))$ is locally the graph of a holomorphic map.*

Proof. Existence of C^∞ solutions to (1,2) follow by the smooth versions of inverse and implicit function theorems, so we prove only holomorphy. ■

Definition 1.1.38 (Laz 2.1). Let X be a smooth manifold.

1. A holomorphic atlas $\mathcal{A} = \{(U_i, \varphi_i)\}_{i \in I}$ consists of an open cover $\{U_i\}_{i \in I}$ together with diffeomorphisms $\varphi : U_i \cong V_i \subseteq \mathbb{C}^n$, n fixed, V_i open, such that the transition functions $\varphi_j \circ \varphi_i^{-1}$ are holomorphic for all $i, j \in I$.
2. Two holomorphic atlases are *compatible* if their union is a holomorphic atlas.
3. A complex manifold of dimension n is a smooth $2n$ -dimensional manifold together with an equivalence class of compatible holomorphic atlases (basically the same as a maximal atlas).

Definition 1.1.39 (Laz 2.2). Let X be a complex manifold.

1. A function $f : X \rightarrow \mathbb{C}$ is *holomorphic* if $F \circ \varphi_i^{-1} : \varphi(U_i) \subseteq \mathbb{C}^n \rightarrow \mathbb{C}$ is holomorphic for every i .
2. If Y is also a complex manifold, then $f : X \rightarrow Y$ is holomorphic if (writing $\{V_j, \psi_j\}_{j \in J}$ for the atlas on Y), for all i, j ,

$$\psi_j \circ F \circ \varphi_i^{-1} : \varphi(U_i) \rightarrow \psi(V_j)$$

is holomorphic $\mathbb{C}^n \rightarrow \mathbb{C}^m$.

Example 1.1.40. All open subsets of complex manifolds are complex manifolds by restricting along the charts. A more interesting example is by viewing S^2 as the one point compactification of $\mathbb{R}^2$ via taking the north pole to be ∞ , and identifying $S^2 \setminus \{N\}$ with $\mathbb{R}^2$ via the stereographic projection. The transition map from the north and south charts is $z \mapsto 1/z$, which is holomorphic.

Now we do analytic hypersurfaces. Let $U \subseteq \mathbb{C}^{n+1}$ open and $f : U \rightarrow \mathbb{C}$ holomorphic. Take $Z(f)$ the zero locus of f inside of U . The implicit function theorem tells us when this is a manifold. Let for all $a \in Z(f)$, the vector $\partial_{z_i} f(a)$ be nonzero for some i . Then by the IFT, $Z(f)$ is locally given as the graph of a holomorphic map $g : (V \subseteq \mathbb{C}^n) \rightarrow Z(f)$; i.e. $Z(f) = \{(w, g(w))\}$. This is then a manifold.

It is important to be careful and ensure that spaces are bounded. For example, the zero locus of $z_1^2 + z_2^2 - 1$ is not bounded.

Chapter 2

Complex manifolds

2.1 Definitions and examples

Lecture 5: Complex manifolds

Tue 03 Feb 2026 12:33

Example 2.1.1. If X, Y are complex manifolds, then $X \times Y$ is a complex manifold since it is a smooth manifold, and products of holomorphic functions are holomorphic by definition.

Let $\{f_i\}_{1 \leq i \leq k}$ holomorphic $X \rightarrow \mathbb{C}$, X a complex manifold. Then we can define the zero locus on *all* functions $Z(f_1, \dots, f_k) \subseteq X$ as

$$\{z \in X \mid f_i(z) = 0 \forall i\}.$$

If

$$J(f_1, \dots, f_k)_j^i = \frac{\partial f_i}{\partial z_j}$$

has full rank at all z , then the zero locus is a complex submanifold of X . We do this by taking the zero locus as a curve-like thingy and straightening it.

Another example is complex Lie groups. These are very simply Lie groups that are complex manifolds with multiplication and inversion holomorphic. Equivalently, it is a group object in $\mathbb{C}\text{plxMan}$. For example, consider $\text{GL}(n, \mathbb{C})$. This is equal to $M_{n \times n}(\mathbb{C}) \setminus Z(\det)$, and since the fiber of 0 is closed, $\text{GL}(n, \mathbb{C})$ is open in $\mathbb{C}^{2n}$. Since multiplication is a polynomial in each entry, it is holomorphic. Inversion is also holomorphic since it can, via Cramer's rule, be realized as a division of polynomials, which is a rational function in the entries in this case, and we are working in the nonvanishing locus of $\det$.

There are many subgroups of GL that are submanifolds. For example, $\text{SL}(n, \mathbb{C})$ the set of all matrices with determinant 1. This is the fiber of 1 under $\det$, it suffices to check that 1 is a regular value of $\det$, meaning the Jacobian is surjective on the tangent spaces on each element of the preimage. It suffices to show $d(\det)_A : T_A \text{GL}(n, \mathbb{C}) \rightarrow T_1 \mathbb{C}$ for all $A \in \text{SL}(n, \mathbb{C})$. We are able to work in the non-complexified tangent space, since that will be surjective precisely when the real Jacobian is surjective (follow directly from writing down the complexified Jacobian). To do this, we take a curve passing through A and show its derivative maps to a nonzero tangent vector. Let $\gamma(t) = (1+t)A$. Then

$$\det(\gamma(t)) = (1+t)^n \det A = (1+t)^n.$$

Thus

$$d(\det)_A(\gamma'(0)) = \left. \frac{d}{dt} \right|_0 \gamma(t) = n \neq 0.$$

Many other matrix Lie groups are defined analogously. However, *the unitary and special unitary groups are NOT complex Lie groups*. This follows by a fact that compact complex Lie groups have to be abelian. This follows by considering conjugation, which you can show is always constant. Then since U , SU are compact and nonabelian, the statement follows.

Another example is complex tori. Let $\Lambda \in \mathbb{C}^n$ be a lattice: a free abelian subgroup of the maximal rank $2n$. Hence this is simply the subgroup $\langle \lambda_i \rangle_{1 \leq i \leq 2n}$ for $\lambda_i \in \mathbb{C}^n$ linearly independent. Now define the group quotient $X_\Lambda := \mathbb{C}^n / \Lambda$. The quotient map is the universal cover of X_Λ , and moreover is a local diffeomorphism. Hence we obtain coordinates on X_Λ by taking a local inverse s on U of π , then defining coordinates on U via $s : U \rightarrow V \subseteq \mathbb{C}^n$. Coordinate transformations are translations – affine maps. Those are clearly holomorphic. We claim there is a diffeomorphism $X_\Lambda \cong (S^1)^{2n}$. NOTE that this is NOT an isomorphism of complex manifolds.

We can construct a matrix $A = (\lambda_1 \ \cdots \ \lambda_{2n})$, which is $\mathbb{R}$ -linear and invertible since the vectors λ_i are linearly independent. Thus the inverse is $\lambda_i \mapsto e_i$. This then descends to $\mathbb{R}^{2n} / \Lambda \cong \mathbb{R}^{2n} / \langle e_i \rangle$. We commute the product and quotient as $\prod_{i=1}^{2n} \mathbb{R} / \mathbb{Z} \cong \prod_{i=1}^{2n} S^1$.

However, not all are biholomorphic. The issue is that the matrix need not be $\mathbb{C}$ -linear. We claim all holomorphic maps $X_\Lambda \rightarrow X_{\Lambda'}$ are induced by some $A \in M_{n \times n}(\mathbb{C})$. The universal cover of both of these is $\mathbb{C}^n$, so if $f : X_\Lambda \cong X_{\Lambda'}$, then there is a lift

$$\begin{array}{ccccc} & & & & \mathbb{C}^n \\ & & & \nearrow F & \downarrow \pi \\ \mathbb{C}^n & \xrightarrow{\pi} & X_\Lambda & \xrightarrow{f} & X_{\Lambda'} \end{array}$$

by precomposing X_Λ with its cover. Since F is holomorphic, consider $g(z) = F(z + \lambda_i) - F(z)$. We can show that $g(z) \in \Lambda'$ for all $z \in \mathbb{C}^n$. Note that Λ' is discrete and $\mathbb{C}^n$ is connected. Thus g is constant for all λ_i . We take the derivative to obtain that the holomorphic derivatives of the lift are periodic with respect to Λ . Hence the values are fully determined within one “box” determined by the lattice. Hence $\partial_{z_j} F$ is bounded and holomorphic, since it is determined on a compact bounded subset. Liouville’s theorem says bounded holomorphic functions on $\mathbb{C}$ are constant. Thus we are constant in each variable, and $\partial_{z_j} F$ is constant. Thus we must have F of the form

$$F(z) = \tilde{A}z + b,$$

$\tilde{A}$ an $n \times n$ matrix and $b \in \mathbb{C}^n$ a constant. Translating by $-b$ we can take $F(z) = \tilde{A}z$. Hence our map is (affine; 0 not always mapped to 0) $\mathbb{C}$ -linear.

Since lattices correspond to invertible $2n \times 2n$ matrices, over $\mathbb{R}$ the space of lattices can be defined with dimension $4n^2$. However $GL(n, \mathbb{C})$ has $\mathbb{R}$ -dimension $2n^2$. Thus there are more lattices than these linear maps, and thus not all can be biholomorphic.

Final example for today are Hopf manifolds. Pick $\lambda \in \mathbb{R}$ with $|\lambda| < 1$. Then $\mathbb{Z}$ acts on $\mathbb{C} \setminus \{0\}$ by $m \cdot (z_1, \dots, z_n) = (\lambda^m z_1, \dots, \lambda^m z_n)$. The action is free and *properly discontinuous*, meaning (for an arbitrary G -space) for any open set U and any $g \in G$, then $g(U) \cap U$ is nonempty iff $g = e$. A Hopf manifold is $\mathbb{C}^n \setminus \{0\}$ quotiented by this action. Interestingly, as a complex manifold, this is isomorphic to $S^{2n-1} \times S^1$.

We will generally say *submanifold* to mean an embedded submanifold, and will specify when submanifolds are immersed explicitly.

Lecture 6: Complex manifolds

Thu 05 Feb 2026 12:28

Remark 2.1.2. I finally have good motivation for why complexifying the tangent space is tensoring with $\mathbb{C}$. Let $v \in T_p M$ and $z = x + iy \in \mathbb{C}$. Then $v \otimes z \in T_p^{\mathbb{C}} M$ is equivalent to, by bilinearity,

$$v \otimes (x + iy) = v \otimes x + v \otimes iy.$$

Of course by $\mathbb{R}$ -linearity of $\otimes_{\mathbb{R}}$, this is equal to

$$= x(v \otimes 1) + y(v \otimes i).$$

We can choose to write $v \otimes 1 = v$ and $v \otimes i = iv$, and this really does behave like normal multiplication since normal multiplication is also bilinear. Thus all elements of $T_p^{\mathbb{C}} M$ can be written of the form $u + iv$ for some $u, v \in T_p M$; in our example above, $u = xv$ and $v = yv$ (abusing notation very badly). This is required to have $\partial_z = \frac{1}{2}(\partial_x - i\partial_y)$ in the tangent space.

Proposition 2.1.3. *Let X be a compact complex connected submanifold, and $f : X \rightarrow \mathbb{C}$ a holomorphic function. Then f is constant.*

Proof. Easy to show f is constant on connected components by passing to coordinates. ■

Corollary 2.1.4. *If $X \subseteq \mathbb{C}^n$ is a compact complex connected submanifold, then X is a point.*

Proof. Take the holomorphic function

$$X \hookrightarrow \mathbb{C}^n \xrightarrow{\pi_i} \mathbb{C}.$$

This must be constant by proposition 3.2.19. ■

Since looking for compact complex connected subsets of $\mathbb{C}^n$, we look to projective space for something like Whitney's theorem for embedding complex manifolds into $\mathbb{C}^n$. This still is not perfect, but is worth considering.

2.2 The projective space

Recall that projective space $k\mathbb{P}^n$ over a field k is defined as the set of lines through the origin in $k^{n+1} \setminus \{0\}$, topologized in the obvious way:

$$k\mathbb{P}^n := \frac{k^{n+1} \setminus \{0\}}{\langle \lambda x \sim x \mid x \in k^{n+1}, \lambda \in k \setminus \{0\} \rangle}.$$

Given π the canonical projection and L any line in $\mathbb{C}^{n+1}$, we have $\pi^{-1}(L) \cong L \setminus \{0\} \cong \mathbb{C}^*$. We can then show that $\mathbb{C}^{n+1} \setminus 0$ is a locally trivial $\mathbb{C}^*$ -bundle over $\mathbb{CP}^n$. Recall that the atlas is formed by sets

$$U_i := \{[c_0 : \cdots : c_n] \in \mathbb{CP}^n \mid c_i \neq 0\}.$$

This is well-defined. The coordinates simply drop c_i , since we can always take it to be normalized at 1.

Remark 2.2.1. We work in the quotient topology on $\mathbb{CP}^n$, which is not the same as the usual topologies we would use in algebraic geometry. However many of the results translate between the two, due to some work by Serre.

We equip $\mathbb{C}^{n+1}$ with the Hermitian inner product $\langle z, w \rangle = \sum_i z_i \bar{w}_i$, the *opposite* of the physics convention.

The fibration

$$S^1 \hookrightarrow S^{n+1} \longrightarrow \mathbb{CP}^n$$

is called the Hopf fibration.

Complex projective space is a CW complex. We can define $\mathbb{CP}^1 := \mathbb{C}^1 \amalg \mathbb{C}^0$ with $\mathbb{C}^0$ the point at ∞ . If we take a chart $U_0 \subseteq \mathbb{CP}^2$, then $\mathbb{CP}^2 \setminus U_0$ is just the set $\{[0 : z_1 : z_2]\}$. Since $U_0 \cong \mathbb{C}^2$, we have $\mathbb{CP}^2 := \mathbb{C}^2 \amalg \mathbb{C}^1 \amalg \mathbb{C}^0$. We can induct to obtain

$$\mathbb{CP}^n := \coprod_{i=0}^n \mathbb{C}^i.$$

Hence $\mathbb{CP}^n$ admits this cell decomposition and is a CW complex with cells of even real dimension. Here we of course have not made the gluing relations explicit. Since cohomology of a CW complex comes by each level of the cells, and every other degree of cells is zero, the cohomology $H^\bullet(\mathbb{CP}^n; \mathbb{C}) \cong \mathbb{C}/0 \cong \mathbb{C}$. Oh and also $\mathbb{CP}^n \cong S^{n+1}/S^1$ a diffeomorphism.

Submanifolds (more generally, subvarieties) of $\mathbb{CP}^n$ admit descriptions as *homogeneous polynomials*. A polynomial f in $n+1$ variables is *homogeneous of degree d* if $f(\lambda x) = \lambda^d f(x)$, $x \in \mathbb{C}^{n+1}$. It follows that f is a sum of monomials of degree d .

Lecture 7: Complex projective space

Tue 10 Feb 2026 12:29

Notation 2.2.2. Given a manifold X and an open set $U \subseteq X$, we will now write $\mathcal{O}_X(U)$ for the set of holomorphic functions $U \rightarrow \mathbb{C}$. Recall by ?? that if X is compact, then $\mathcal{O}_X(X) \cong \mathbb{C}$.

Let f be a homogeneous polynomial of degree d . It makes sense to consider the vanishing locus $Z(f)$ as a subset of $\mathbb{CP}^n$ since if f vanishes on z , then it vanishes on $\{\lambda z\}_{\lambda \in \mathbb{C}} = [z]$.

Definition 2.2.3. Let $F_1, \dots, F_r$ be homogeneous polynomial of degree d in $\mathbb{C}[z_0, \dots, z_n]$. Then the zero locus is defined as

$$Z(F_1, \dots, F_r) = \{[z] \in \mathbb{CP}^n \mid F_i([z]) = 0 \forall i\}.$$

Subsets of this are called *projective varieties*.

Note that any complex (embedded) manifold of $\mathbb{CP}^n$ is compact.

Theorem 2.2.4 (Chow). *Any complex submanifold of $\mathbb{CP}^n$ is cut out by homogeneous polynomials. In particular, all complex submanifolds of $\mathbb{CP}^n$ are algebraic.*

Proposition 2.2.5. *Given F a homogeneous polynomial in $\mathbb{C}[z_0, \dots, z_n]$, suppose that $\partial_{z_i} F$ has no common zeroes in $\mathbb{CP}^n$ for all i . Then $Z(F)$ is a manifold.*

Proof. Let F be homogeneous of degree d . There is an identity known as *Euler's identity* which says

$$\sum_{i=0}^n z_i \frac{\partial F}{\partial z_i} = dF(z).$$

Here d is the degree, NOT the exterior derivative (it is my convention to write that as d). We prove this by differentiating $F(\lambda z) = \lambda^d F(z)$ on both sides with respect to λ and applying the chain rule, then setting $\lambda = 1$.

We want to verify that $Z(F) \cap U_i$ is a manifold. For simplicity take $i = 0$ without loss of generality. Take $[a] \in Z(F) \cap U_i$; we then may freely set $a_i = 1$. Using the identity,

$$\left. \frac{\partial F}{\partial z_0} + a_1 \frac{\partial F}{\partial z_1} + \cdots + a_n \frac{\partial F}{\partial z_n} \right|_{(1, a_1, \dots, a_n)} = dF(1, a_1, \dots, a_n).$$

The right side vanishes. Hence $Z(F) \cap U_0 = Z(F(1, z_1, \dots, z_n))$. To check that this is a manifold using the implicit function theorem, we need a nonzero partial derivative, and thus the Jacobian will have rank 1. The hypothesis and non-zero-ness of $[a]$ guarantees this. ■

Let $X \subseteq \mathbb{CP}^n$ a manifold or projective variety, and $\{F_i\}_{0 \leq i \leq s} \subseteq \mathbb{C}[z_0, \dots, z_n]$ all homogeneous of the same degree d . We claim the map $f : X \rightarrow \mathbb{CP}^s$ defined by $[x] \mapsto [F_0(x) : \dots : F_s(x)]$ is well-defined. This follows by homogeneity of the F s, *except* for the fact that $f[x]$ needs to be nonzero everywhere. This happens if and only if $X \cap Z(F_0, \dots, F_s)$ is empty. We claim f is holomorphic. Let $x \in U_j$ and $f(x) \in U_k$, then in coordinates

$$f(x) = \left\{ \frac{F_i(z_0, \dots, \hat{z}_j, \dots, z_n)}{F_k(z)} \right\}_{0 \leq i \leq s}.$$

This is holomorphic. A fascinating result we will see later is that *all* holomorphic maps arise of this form.

Recall that $\mathrm{GL}(n+1, \mathbb{C})$ acts on $\mathbb{C}^{n+1}$, and this descends to $\mathbb{CP}^n$. However this is not a faithful action, since some matrices act trivially: anything that scales stuff. This is the center, so we can mod out by the center $\mathrm{GL}(n+1, \mathbb{C})/\mathbb{Z}(\mathrm{GL}(n+1, \mathbb{C}))$, and this acts faithfully. We claim that these are precisely the automorphisms of $\mathbb{CP}^n$. We may prove this later.

Let $X \subseteq \mathbb{CP}^r$ and $Y \subseteq \mathbb{CP}^s$ are manifolds. Any submanifold of projective space will be called a projective manifold. We know $X \times Y$ is a complex manifold, but we do not yet know if it is projective. We proceed with the following proposition; $X \times Y$ will then be projective by a simple corollary.

Proposition 2.2.6 (Segre's embedding). $\mathbb{CP}^r \times \mathbb{CP}^s$ is a submanifold of $\mathbb{CP}^{rs+r+s}$.

Proof. Think of $\mathbb{C}^{rs+r+s+1} = \mathbb{C}^{(r+1)(s+1)}$ as matrices of size $(r+1) \times (s+1)$ over $\mathbb{C}$. Then $\mathbb{CP}^{rs+r+s}$ is just $\mathbb{P}(M_{(r+1) \times (s+1)}(\mathbb{C}))$. Consider the map $\mu_{r,s} : \mathbb{CP}^r \times \mathbb{CP}^s \rightarrow \mathbb{P}(M_{(r+1) \times (s+1)}(\mathbb{C}))$ defined by

$$([a], [b]) \mapsto [c_{ij}], \quad c_{ij} = a_i b_j.$$

For example,

$$([a_0 : a_1], [b_0 : b_1]) \mapsto \left[\begin{pmatrix} a_0 b_0 & a_0 b_1 \\ a_1 b_0 & a_1 b_1 \end{pmatrix} \right],$$

where of course the class is defined by $c_{ij} \sim \lambda c_{ij}$. We claim the image of this map is precisely (classes of) all matrices of rank 1 (all matrices in a class of a matrix with rank 1 will have rank 1). The argument is messy so we do it for our 2 dimensional example. We can take a_0 out of the first row and a_1 out of the second row to get

$$\begin{pmatrix} b_0 & b_1 \\ b_0 & b_1 \end{pmatrix}.$$

This has at most rank 1. Since $a, b \neq 0$ it has at least rank 1. In this example, this is a submanifold since it is precisely the fiber of 0 under $\det$. ■

Lecture 8: Toric geometry

Thu 19 Feb 2026 12:22

Class taught by Lau today. Consider $\mathbb{C}^\times = \mathbb{C} \setminus \{0\}$. Topologically this is a cylinder. We have a map $\mathbb{C}^\times \rightarrow \mathbb{R}$ via $z \mapsto \log |z|$. Note that S^1 acts on $\mathbb{C}^\times$ by multiplication, and our map $z \mapsto \ln |z|$ is S^1 -equivariant. Now consider taking the dimension to n :

$$(\mathbb{C}^\times)^n \rightarrow \mathbb{R}^n, \quad \{z_i\}_{1 \leq i \leq n} \mapsto \{\ln |z_i|\}_{1 \leq i \leq n}.$$

Then the n -torus $T^n = (S^1)^n$ acts on this in the same way, and our map is T^n -equivariant.

We might want to allow this to be $\mathbb{C}$ instead of $\mathbb{C}^\times$, but $\ln 0 = -\infty$, so we need to take our map $\mathbb{C} \rightarrow \mathbb{R} \cup \{-\infty\}$. We then think of $\mathbb{C}$ as kind of like a cylinder that is infinite in one dimension and pinched off at 0 in the other, since each circle is identified down by taking $\ln$ of its modulus. We could then try something like

$$\mathbb{C}^n \times (\mathbb{C}^\times)^m \rightarrow (\mathbb{R} \cup \{-\infty\})^n \times \mathbb{R}^m$$

defined in the same T^{n+m} -equivariant way.

Consider $\mathbb{CP}^1$. We can apply our toric point of view here as well. Consider the map $\mathbb{CP}^1 \rightarrow \mathbb{R}$ in coordinates as follows. In U_0 , we map $[z_0 : z_1] = [1, z_1/z_0] \mapsto \log |z_1/z_0|$. We do the same for U_1 . We can do a similar construction for $\mathbb{CP}^2 \rightarrow \mathbb{R}^2$ by dividing out by the nonzero element of a chart and applying our construction on each coordinate. This allows us to visualize $\mathbb{CP}^n$ as polytopes and stuff in $\mathbb{R}^n$ modulo a toric action. It also makes it much easier to read off T^n -equivariant morphisms of complex manifolds.

A *toric variety* is a n -variety X that admits a $(\mathbb{C}^\times)^n$ -action with an open dense orbit. We then take the T^n -subaction. Importantly, we can also reverse this process and obtain a manifold back by taking the dual cone of a cone and then just constructing back our manifold.

2.3 Holomorphic vector bundles

Lecture 9: Holomorphic vector bundles

Thu 26 Feb 2026 12:31

Consider a map $X \rightarrow \mathbb{CP}^m$ for $X \subseteq \mathbb{CP}^n$ a manifold. Such a map is given by a collection $\{p_i\}_{0 \leq i \leq m}$ of homogeneous polynomials of the same degree that do not simultaneously vanish on X . We will see that this can be described as follows: maps $X \rightarrow \mathbb{CP}^m$ are given by a line bundle $\mathcal{L} \rightarrow X$ together with a collection of holomorphic sections that don't simultaneously vanish. To understand this, we define holomorphic vector bundles.

There are two approaches to holomorphic vector bundles. We start with the *total space picture*, and show how this is equivalent to the *cocycle picture*.

Definition 2.3.1. Let X be a complex manifold. A holomorphic vector bundle E over X of rank k is a complex manifold E together with a holomorphic map $\pi : E \rightarrow X$ such that

- there exists an open cover $\{U_i\}_{i \in I}$ of X and biholomorphisms $\varphi_i : \pi^{-1}(U_i) \rightarrow U_i \times \mathbb{C}^k$ making the following diagram commute:

$$\begin{array}{ccc} \pi^{-1}(U_i) & \xrightarrow{\varphi_i} & U_i \times \mathbb{C}^k \\ \pi \searrow & & \swarrow \pi_i \\ & U_i & \end{array}$$

- Write $U_{ij} = U_i \cap U_j$. Then the transition functions $\varphi_{ij} := \varphi_i \circ \varphi_j^{-1} : U_{ij} \cap \mathbb{C}^k \rightarrow U_{ij} \cap \mathbb{C}^k$ must be biholomorphic and of the form

$$\varphi_{ij}(x, v) = (x, g_{ij}(x)v)$$

for some $g_{ij} \in \text{GL}(k, \mathcal{O}_X(U_{ij}))$; i.e. g_{ij} is a biholomorphic map $U_{ij} \rightarrow \text{GL}(k, \mathbb{C})$.

Note that $g_{ij} = g_{ji}^{-1}$, and on a triple overlap U_{ijk} , we have $g_{ij} \cdot g_{jk} = g_{ik}$. This is called the *cocycle condition*, and the collection $\{g_{ij}\}_{i,j \in I}$ is called the *cocycle* of E .

Terminology 2.3.2. A holomorphic vector bundle of rank 1 is called a holomorphic *line bundle*.

Example 2.3.3. A simple example is the trivial bundle $X \times \mathbb{C}^k \rightarrow X$. For a more interesting example, consider $\mathcal{O}_{\mathbb{CP}^n}(-1)$. Given $[a] \in \mathbb{CP}^n$, write L for the line $\{([a], v) \mid v \in \mathbb{C}a\}$. We project by $p|L : L \rightarrow \mathbb{CP}^n$. We use the usual cover $\{U_i\}_{0 \leq i \leq n}$ where the i th homogeneous coordinate is nonzero in U_i , and its preimage $p^{-1}(U_i)$ is $U_i \times \{v \in \mathbb{C}^{n+1} \mid v \in [z] \in U_i\}$. We then write $v = t[z]$ for the representative of $[z]$ with $z_i = 1$, and map $([z], v) \mapsto ([z], t)$. This gives a map $p^{-1}(U_i) \cong U_i \times \mathbb{C}$. By construction the relevant diagram commutes. Since $\text{GL}(1, \mathcal{O}_X(U_{ij})) \cong \mathcal{O}_X(U_{ij})$, we compute the cocycle by finding a holomorphic function which is nonvanishing on the overlaps. Define

$$g_{ij} : U_{ij} \rightarrow \text{GL}(1, \mathbb{C}) = \mathbb{C}^*$$

as follows. Define

$$g_{ij} : U_{ij} \times \mathbb{C} \rightarrow U_{ij} \times \mathbb{C}$$

as follows. View the first U_{ij} as $\subseteq U_j$, and the second as U_i . By this we mean set $z_j = 1$ in the first, and $z_i = 1$ in the second. The transition function φ_{ij} is $\varphi_i \circ \varphi_j^{-1}$, which maps

$$\left(\left[\frac{z_0}{z_j} : \cdots : 1 : \cdots : \frac{z_n}{z_j} \right], t \right) \mapsto \left(\left[\frac{z_0}{z_j} : \cdots : 1 : \cdots : \frac{z_n}{z_j} \right], t[z] \right) = \left(\frac{z_j}{z_i}[z], t \frac{z_j}{z_i}[z] \right) \mapsto \left([z], \frac{z_j}{z_i} t \right).$$

Hence $g_{ij} = \frac{z_j}{z_i}$.

It will be useful to see how cocycles give rise to holomorphic vector bundles, since it is easier to see if cocycles are cohomologous in some sense, rather than constructing isomorphisms on the total space.

Suppose we are given, for all i, j , a cover $\{U_i\}_{i \in I}$ and maps $g_{ij} : U_{ij} \rightarrow \text{GL}(k, \mathbb{C})$ satisfying

- $g_{ij} = g_{ji}^{-1}$ and
- $g_{ij} \cdot g_{jk} = g_{ik}$.

Then we can build the total space and projection of a vector bundle of rank k $p : E \rightarrow X$ as follows. We start by taking the disjoint union of the covers, and then glue along the relation

$$E := \left(\coprod_{i \in I} U_i \times \mathbb{C}^k \right) / \langle (x, v) \sim (x, g_{ij}(v)) \forall x \in U_{ij} \rangle.$$

There is some bookkeeping needed to check E has a complex manifold structure. To define $p : E \rightarrow X$, choose $(x, v) \in [x, v]$, note that x is fixed, and map $p[x, v] = x$. Because the equivalence relation we glued along is linear, we find that all relevant universals for vector spaces can be done to vector bundles.

We compute the cocycles for the dual E^* for fun. Let $p : E \rightarrow X$ be a vector bundle of rank k with local trivializations $\varphi_i : p^{-1}(U_i) \rightarrow U_i \times \mathbb{C}^k$. Write $\varphi_i \circ \varphi_j^{-1} = g_{ij}$. When we dualize, the map g_{ij}^t goes the opposite direction, so we invert it to obtain $(g_{ij}^t)^{-1}$ as a cocycle for E^* . Proving this is a cocycle is like effort.

Let $E, F \rightarrow X$ be represented by $\{g_{ij}\}, \{h_{rs}\}$. WLOG assume they are trivialized on the same open cover, since otherwise we may refine it. We represent $E \oplus F$ by $g_{ij} \oplus h_{ij}$ (diagonal block matrix). We represent $E \otimes F$ by $g_{ij} \otimes h_{ij}$. We represent the *highest* exterior power $\Lambda^k E$, which should have rank 1, by $\{\det g_{ij}\}$.

Definition 2.3.4. Let $p : E \rightarrow X$ be a holomorphic vector bundle. A (global) holomorphic section of E is a holomorphic map $s : X \rightarrow E$ such that $p \circ s = 1$. We write $\Gamma(E)$ or $\Gamma(X, E)$ for the set of global sections of E .

One defines local sections in the same way. Note here that any holomorphic vector bundle E on X defines a sheaf on X , also denoted by E , by taking sections of the sheaf E to be sections of the vector bundle E .

If $\varphi_i : p^{-1}(U_i) \rightarrow U_i \times \mathbb{C}^k$ is a trivialization, then $\varphi_i \circ s|_{U_i} : U_i \rightarrow U_i \times \mathbb{C}^k$ is a vector-valued function $x \mapsto (x, (\varphi_i \circ s)(x))$. Here we abuse notation and write $\varphi(p) = (p, \varphi(p))$ since φ must act in this way. We also note that, writing $\varphi_i \circ s = f_i$, we have $f_i = g_{ij} f_j$.

Consider $\mathcal{O}_{\mathbb{CP}^1}(-1)$ and its dual $\mathcal{O}_{\mathbb{CP}^1}(1)$ on $\mathbb{CP}^1$. Represent these by g, h . Then

$$g_{01} = \frac{z_0}{z_1}, \quad h_{01} = \frac{z_1}{z_0} = \frac{1}{g_{01}}.$$

I got this backwards earlier. A section of $\mathcal{O}_{\mathbb{CP}^1}(-1)$ is a pair $f_0, f_1 : \mathbb{C} \rightarrow \mathbb{C}$ (which look like polynomials $a_0 + a_1 t + \dots$ and $b_0 + b_1 t^{-1} + b_2 t^{-2} + \dots$), such that $f_0 = \frac{1}{t} f_1$ (agree under g_{ij} on overlap). There is no way for this to happen unless $f_0 = f_1 = 0$. Thus $\mathcal{O}_{\mathbb{P}^1}(-1)$ has no nontrivial sections, so $\Gamma(\mathbb{P}^1, \mathcal{O}_{\mathbb{P}^1}(-1)) \cong 0$. The same will not be true of the dual.

Lecture 10: Continuing holomorphic vector bundles

Tue 03 Mar 2026 12:17

We remark that $\mathcal{O}_{\mathbb{P}^n}(-1)$ is basically $\mathbb{P}^n \times \mathbb{C}^{n+1}$ apparently.

We then define

$$\mathcal{O}_{\mathbb{P}^n}(k) := \mathcal{O}_{\mathbb{P}^1}^{\otimes k},$$

and the corresponding cocycles are given by the tensors of the cocycles. Since the line bundles have dimension 1, the k -fold tensor of $g_{ij} = z_j/z_i$ is $g_{ij}^{\otimes k} = z_j^k/z_i^k$. Anyways the space of global sections $\Gamma(\mathbb{P}^n, \mathcal{O}_{\mathbb{P}^n}(k))$ is isomorphic to the homogeneous polynomials in $\{z_0, \dots, z_k\}$ of degree k . Recall of course that homogeneous polynomials are precisely polynomials in the coordinates of $\mathbb{P}^n$. In particular, the space is finite-dimensional.

Remark 2.3.5. We will see later, using Dolbeaux cohomology, that this generalizes: if $p : E \rightarrow X$ is a holomorphic vector bundle over a compact manifold has a finite dimensional $\mathbb{C}$ -vector space of global sections.

Example 2.3.6. Let X be a complex manifold with atlas $\{(U_i, \varphi_i)\}_{i \in I}$. Write $\psi_{ij} = \varphi_i \circ \varphi_j^{-1}$. Let $g_{ij} = J(\psi_{ij})$ for J the holomorphic Jacobian. This is a cocycle. Recall that taking the Jacobian preserves composition. Also note that $\psi_{ij} \circ \psi_{jk} = \psi_{ik}$, and thus $J(\psi_{ij} \circ \psi_{jk}) = J(\psi_{ij}) \cdot J(\psi_{jk}) = J(\psi_{ik})$. Additionally we have $J(f \circ f^{-1}) = J(1) = 1$ for any $f \in C^\infty(U)$, so the cocycle conditions are satisfied. We call the bundle this induces the holomorphic cotangent bundle $\mathcal{T}_X^*$, which will be the set of 1-forms $\Omega_{\text{holo}}^1(X)$. We will discuss the linear algebra of the total space construction later.

Remark 2.3.7. Let $\mathcal{L}_1, \mathcal{L}_2$ be line bundles on X . Then $\mathcal{L}_1 \otimes \mathcal{L}_2$ is also a line bundle. Additionally, $\Gamma(\mathcal{L}_1 \otimes \mathcal{L}_1^*, -) \cong \mathcal{O}_X$. To see this, if $\{g_{ij}\}$ represents $\mathcal{L}_1$ then $\{\frac{1}{g_{ij}}\}$ represents $\mathcal{L}_1^*$. Hence their product is 1. Alternatively, note that $\mathcal{L}_1 \otimes \mathcal{L}_1^*$ corresponds to endomorphisms of $\mathcal{L}_1$. Consider the identity and play with that.

Definition 2.3.8. Isomorphism classes of line bundles on a complex manifold X form a group of line bundles known as the *Picard group*, denoted $\text{Pic}(X)$.

We will see using more advanced machinery that $\text{Pic}(\mathbb{P}^n) \cong \mathbb{Z}^n$, a highly unusual fact following since $\mathbb{P}^n$ is some special type of variety.

Exercise 2.3.9. Let $p : E \rightarrow X$ be a holomorphic bundle. There is a correspondence between trivializations of a bundle $E|U \rightarrow U \times \mathbb{C}^r$ and holomorphic frames on U .

Notation 2.3.10. Note that if $p : E \rightarrow X$ is any vector bundle and $x \in X$, then $p^{-1}(x) =: E_x$ is a vector space by choosing a trivialization.

Definition 2.3.11. Let $p_1 : E \rightarrow X$ and $p_2 : F \rightarrow X$ be holomorphic vector bundles on a complex manifold X . A homomorphism $\psi : E \rightarrow F$ is a holomorphic map such that

1. $\psi|E_x : E_x \rightarrow F_x$ is linear, and
2. $\text{rk} \psi_x$ is constant as a function over $x \in X$, and
3. the diagram

$$\begin{array}{ccc} E & \xrightarrow{\psi} & F \\ & \searrow p_1 & \swarrow p_2 \\ & X & \end{array}$$

commutes.

We can construct kernels and cokernels of maps in the usual way, hence the category of holomorphic vector bundles is an abelian category.

We want to recast this definition in terms of cocycles. We start with trivial bundles. A homomorphism $\psi : E \rightarrow F$ gives us local maps $\psi_i : p^{-1}(U_i) \cong U_i \times \mathbb{C}^e \rightarrow p^{-1}(U_i) \cong U_i \times \mathbb{C}^f$ via $(x, v) \mapsto (x, \psi_i(x) \cdot v)$. Hence ψ_i is an $f \times e$ matrix of holomorphic functions. So we end up with a diagram

$$\begin{array}{ccc} p_1^{-1}(U_j) & \xrightarrow{\psi_j} & p_2^{-1}(U_j) \\ g_{ij} \downarrow & & \downarrow h_{ij} \\ p_1^{-1}(U_i) & \xrightarrow{\psi_i} & p_2^{-1}(U_i) \end{array}$$

Hence the general way to write a cocycle is to have matrices of holomorphic functions $\{\psi_i\}$ which are constant rank, and make this diagram commute.

Remark 2.3.12. Define E as a category with objects all U_i in the cover, and precisely one morphism $U_i \rightarrow U_j$ for all i, j . Hence this is a groupoid. We may then consider vector bundles as a functor, and morphisms of vector bundles are natural transformations between them. This is a somewhat naïve picture and there are indeed issues with trying to generalize this.

The correspondence of cocycles with holomorphic vector bundles is not precise, and is only bijective up to some equivalence relation on cocycles. We will explore this today, but the idea is that various open covers can admit the same cocycles.

Chapter 3

Differential calculus on complex manifolds

3.1 Complex and Hermitian structures

Lecture 11: Complex and hermitian linear algebra

Tue 17 Mar 2026 12:30

In this section, we explore indepth the implications of complexification of a real vector space. This involves working with almost complex vector spaces.

Definition 3.1.1. Let V be a $\mathbb{R}$ -vector space. An *almost complex structure* on V is an $\mathbb{R}$ -linear endomorphism $J : V \rightarrow V$ such that $J^2 = -1$.

Note that J has eigenvalues $\pm i$ so it cannot be $\mathbb{R}$ -diagonalized. Note the following fairly obvious statement:

Lemma 3.1.2. Any almost complex vector space (V, J) admits the structure of a complex vector space in a natural way.

Proof. Define $(a + ib)v = av + bJ(v)$. This is a $\mathbb{C}$ -bilinear map since $(i + i)(v) = (J + J)(v) = J(v) + J(v)$. ■

Definition 3.1.3. Let V be a $\mathbb{R}$ -vector space. The *complexification* of V , denoted $V_{\mathbb{C}}$, is the tensor product $V_{\mathbb{C}} := V \otimes_{\mathbb{R}} \mathbb{C}$.

Remark 3.1.4. There is a canonical inclusion $V \hookrightarrow V_{\mathbb{C}}$ via the map $v \mapsto v \otimes 1$. Hence we may write $V \subseteq V_{\mathbb{C}}$. We write $v \otimes 1$ as v and $v \otimes i$ as iv for any $v \in V$, and define complex conjugation $(v \otimes \lambda) = v \otimes \bar{\lambda}$. Then $V \subseteq V_{\mathbb{C}}$ is precisely the part that is invariant under complex conjugation. We can also write all $v \in V_{\mathbb{C}}$ as $v = x + iy$ with notation as before, and $\bar{v} = x - iy$.

Notation 3.1.5. If (V, J) is an almost complex vector space, then we also denote by $V_{\mathbb{C}}$ the extension of V to a $\mathbb{C}$ -vector space as defined in lemma 3.1.2. These notations are immediately reconciled by taking the almost complex structure on V to be multiplication by i . We also denote by J the extension of J to a $\mathbb{C}$ -linear endomorphism $V_{\mathbb{C}} \rightarrow V_{\mathbb{C}}$ defined by $J(i) = iJ(1)$.

Definition 3.1.6. Let (V, J) be an almost complex vector space, and also write $J : V_{\mathbb{C}} \rightarrow V_{\mathbb{C}}$ as in ?? 3.1.5. Then we define $V^{1,0}$ to be the i -eigenspace of J , and $V^{0,1}$ to be the $-i$ -eigenspace of J . More precisely,

$$V^{1,0} = \{v \in V_{\mathbb{C}} \mid J(v) = iv\}, \quad V^{0,1} = \{v \in V_{\mathbb{C}} \mid J(v) = -iv\}.$$

Remark 3.1.7. Let $v + iw \in V^{1,0}$. Here we *do not assume* $v, w \in V$. Then

$$J(v + iw) = i(v + iw) = iv - w.$$

Since J is $\mathbb{C}$ -linear, we obtain $J(v) = -w$ and $J(w) = v$. We obtain

$$V^{1,0} = \{v - iJ(v) \mid v \in V_{\mathbb{C}}\}.$$

Similarly,

$$V^{0,1} = \{v + iJ(v) \mid v \in V_{\mathbb{C}}\}.$$

Lemma 3.1.8. *Let (V, J) be an almost complex vector space. Then*

$$V_{\mathbb{C}} \cong V^{1,0} \oplus V^{0,1}.$$

Proof. Note that $V^{1,0} \cap V^{0,1} = 0$, since $-iv = J(v) = iv$ would imply $v = 0$. It suffices to write any $v + iw \in V_{\mathbb{C}}$ as a linear combination of elements in $V^{1,0}$ and $V^{0,1}$. Indeed, write

$$v = \frac{1}{2}(v - iJ(v)) + \frac{1}{2}(v + iJ(v)).$$

■

Discussion 3.1.9. We collect a corollary of lemma 3.1.8. There are canonical projections $\pi^{1,0} : V^{1,0} \rightarrow V_{\mathbb{C}}$ and $\pi^{0,1} : V^{0,1} \rightarrow V_{\mathbb{C}}$ defined by

$$\begin{aligned} \pi^{1,0}(w) &= \pi^{1,0} \left(\frac{1}{2}(w - iJw) + \frac{1}{2}(w + iJw) \right) = \frac{1}{2}(w - iJw), \\ \pi^{0,1}(w) &= \frac{1}{2}(w + iJw). \end{aligned}$$

Lemma 3.1.10. *Let (V, J) be an almost complex vector space. Then the real dual $V^* = \text{Hom}_{\mathbb{R}}(V, \mathbb{R})$ has a natural almost complex structure given by $J(f)(v) = f(J(v))$. Then there is an isomorphism $(V_{\mathbb{C}})^* \cong (V^*)_{\mathbb{C}}$ which decomposes as*

$$\begin{aligned} (V^*)^{1,0} &= \{f \in \text{Hom}_{\mathbb{R}}(V, \mathbb{C}) \mid f(I(v)) = if(v)\} = (V^{1,0})^* \\ (V^*)^{0,1} &= \{f \in \text{Hom}_{\mathbb{R}}(V, \mathbb{C}) \mid f(I(v)) = -if(v)\} = (V^{0,1})^*. \end{aligned}$$

Also note that $(V^*)^{1,0} = \text{Hom}_{\mathbb{C}}((V, J), \mathbb{C})$.

Remark 3.1.11. Let (V, J) be an almost complex vector space. The composite

$$(V, J) \hookrightarrow V_{\mathbb{C}} \xrightarrow{\pi^{1,0}} V^{1,0}$$

is an isomorphism of $\mathbb{C}$ -vector spaces.

Discussion 3.1.12. Let (V, J) be an almost complex vector space. Let $\{z_i = \frac{1}{2}(x_i - iy_i)\}_{1 \leq i \leq n}$ be a $\mathbb{C}$ -basis for $V^{1,0}$. Then we find $y_i = J(x_i)$ and that $\{x_i, y_i\}_{1 \leq i \leq n}$ forms a $\mathbb{R}$ -basis for V , and thus a $\mathbb{C}$ -basis of $V_{\mathbb{C}}$. A natural basis for $V^{0,1}$ is then given by $\bar{z}_i = \frac{1}{2}(x_i + iy_i)$. The dual basis for $(V^*)^{1,0}$ is given by $z^i = \frac{1}{2}(x^i + iy^i)$, where

$$x^i(x_j) = \delta_{ij}, \quad y^i(y_j) = \delta_{ij}, \quad x^i(y_j) = y^i(x_j) = 0$$

are the elements of the dual $\mathbb{C}$ -basis for $V_{\mathbb{C}}$. Additionally, $z^i(z_j) = \delta_{ij}$. We similarly have $\bar{z}^i = \frac{1}{2}(x^i - iy^i)$ and $\bar{z}^i(\bar{z}_j) = \delta_{ij}$ and the other expected identities.

Discussion 3.1.13. Now consider the $(\mathbb{C})$ -exterior algebra of the complexification of an almost complex vector space (V, J) . By lemma 3.1.8,

$$\Lambda^k V_{\mathbb{C}} \cong \Lambda^k (V^{1,0} \oplus V^{0,1}).$$

One notes that this should be generated by wedges $z_1 \wedge \cdots \wedge z_k$, where $z_i = x_i + y_i$ for $x_i \in V^{1,0}$ and $y_i \in V^{0,1}$. Hence it decomposes as a sum of wedges, all of length k , which contain elements of $V^{1,0}$ and $V^{0,1}$, meaning that we should be able to generate the k th exterior power with elements of the form

$$x_1 \wedge \cdots \wedge x_j \wedge y_{j+1} \wedge \cdots \wedge y_k, \quad x_i \in V^{1,0}, \quad y_i \in V^{0,1}.$$

Generally authors define $\Lambda^{p,q} V_{\mathbb{C}} := \Lambda^p V^{1,0} \otimes_{\mathbb{C}} \Lambda^q V^{0,1}$, noting that for some reason the tensor product here is isomorphic to the wedge product via the canonical map $x \otimes y \mapsto x \wedge y$. (This works because we want $2x \wedge y = x \otimes y - y \otimes x$, but $y \otimes x$ doesn't exist in this space because this is a tensor product of disjoint spaces). I think this is all confusing and pointless so I just define it as follows:

$$\Lambda^{p,q} V_{\mathbb{C}} := \Lambda^p V^{1,0} \wedge \Lambda^q V^{0,1} \cong \Lambda^p V^{1,0} \otimes_{\mathbb{C}} \Lambda^q V^{0,1}.$$

We then have

$$\Lambda^k V_{\mathbb{C}} \cong \bigoplus_{0 \leq p, q \leq n, p+q=k} \Lambda^{p,q} V_{\mathbb{C}}.$$

An element $x \in \Lambda^{p,q} V_{\mathbb{C}}$ is said to be of type (p, q) or to have bidegree (p, q) . All exterior products are taken as complex exterior products, meaning wedges are $\mathbb{C}$ -linear.

Remark 3.1.14. Let (V, J) be an almost complex vector space. Given a basis z_i for $V^{1,0}$, then a basis for $\Lambda^{p,q} V_{\mathbb{C}}$ is

$$\{z_{r_1} \wedge \cdots \wedge z_{r_p} \wedge \bar{z}_{s_1} \wedge \cdots \wedge \bar{z}_{s_q}\}$$

where $r_i < r_j$ iff $i < j$ and the same for s . Additionally, one defines

$$\Lambda^k J : \Lambda^k V_{\mathbb{C}} \rightarrow \Lambda^k V_{\mathbb{C}}$$

by

$$(\Lambda^k J)(v_1 \wedge \cdots \wedge v_k) = Jv_1 \wedge \cdots \wedge Jv_k.$$

Definition 3.1.15. Let (V, J) be an almost complex vector space. Define the projections

$$\Pi^{p,q} : \Lambda^{\bullet} V_{\mathbb{C}} \rightarrow \Lambda^{p,q} V.$$

Note that if $x \notin \Lambda^{p,q} V$, then $\Pi^{p,q}(x) = 0$. Similarly, define

$$\Pi^k : \Lambda^{\bullet} V_{\mathbb{C}} \rightarrow \Lambda^k V_{\mathbb{C}}.$$

Now inner products. Every real positive definite inner product on a real vector space is diagonalizable to the identity, so its signature is 1. Up to changes of basis, there is only one such inner product.

Definition 3.1.16. Let (V, J) be an almost complex vector space where $(V, \langle -, - \rangle)$ is an inner product space. Then J is *compatible* with the inner product if it is an *isometry*:

$$\langle J(-), J(-) \rangle = \langle -, - \rangle.$$

An almost complex vector space with compatible inner product is called an *almost complex inner product space*.

Remark 3.1.17. If J is compatible with the inner product, then there is a basis $\{x_1, Jx_1, \dots, x_n, Jx_n\}$, and this is the orientation determined by J . Then J preserves this orientation.

Definition 3.1.18. Let $(V, J, \langle -, - \rangle)$ be an almost complex inner product space. The *fundamental form* ω of V is the form $\omega = \langle J(-), - \rangle$.

Lemma 3.1.19. *The fundamental form is a form. In particular, ω is of type $(1, 1)$.*

Proof. We must show ω is skew-symmetric. This follows:

$$\omega(v, w) = \langle Jv, w \rangle = \langle w, Jv \rangle = \langle Jw, -v \rangle = -\langle Jw, v \rangle = -\omega(w, v).$$

For the second claim, recall that

$$\Lambda^2 V_{\mathbb{C}}^* \cong \Lambda^{2,0} V_{\mathbb{C}}^* \oplus \Lambda^{1,1} V_{\mathbb{C}}^* \oplus \Lambda^{0,2} V_{\mathbb{C}}^*.$$

Since $V^{1,0}$ is the i -eigenspace for J , we obtain that for any $\alpha \in \Lambda^{2,0} V_{\mathbb{C}}^*$, by remark 3.1.14, $(\Lambda^2 J^*)\alpha = -\alpha$. The same holds if $\alpha \in \Lambda^{0,2} V_{\mathbb{C}}^*$. However, we have

$$(\Lambda^2 J^*)(\omega)(v, w) = \omega(Jv, Jw) = \langle J^2 v, Jw \rangle = \langle -v, Jw \rangle = \langle -Jv, J^2 w \rangle = \langle Jv, w \rangle = \omega(v, w).$$

Here we have used the fact that J is compatible with $\langle -, - \rangle$. Therefore $\omega \in \Lambda^{1,1} V_{\mathbb{C}}^*$. ■

Definition 3.1.20. A *Hermitian form* on a complex vector space U is a 2-form H with $H(v, w) = \overline{H(w, v)}$ and $H(\alpha u, \beta v) = \alpha \overline{\beta} H(u, v)$.

Example 3.1.21. Let $(V, J, \langle -, - \rangle)$ be a complex inner product space with fundamental form ω . Define

$$h := \langle \cdot, \cdot \rangle - i\omega.$$

Here we use lemma 3.1.2 to make sense of multiplication by i .

Choose a $\mathbb{C}$ -basis z_i and write $z_i = \frac{1}{2}(x_i - iy_i)$ for $V_{\mathbb{C}}^*$. Then define

$$\langle z_i, z_j \rangle := \frac{1}{2} h_{ij}.$$

As always, the dual basis is $z^i = \frac{1}{2}(x^i + iy^i)$. Then we can compute

$$\omega = \frac{i}{2} \sum_{i,j} h_{ij} z^i \wedge \bar{z}^j.$$

Example 3.1.22. Let $(V, J, \langle -, - \rangle)$ be a real inner product space. One then defines the hermitian inner product on $V_{\mathbb{C}}$ by extending the real inner product:

$$\langle v \otimes \lambda, w \otimes \mu \rangle_{\mathbb{C}} := (\lambda \bar{\mu}) \langle v, w \rangle.$$

A computation that I do not want to type up shows that this is the same as, if $\lambda, \mu \in \mathbb{C}$ and $v, w \in V$, (since the inner product should be $\mathbb{C}$ -sorta-linear):

$$\langle \lambda v, \mu w \rangle = \lambda \bar{\mu} \langle v, w \rangle.$$

In the basis $z^i = \frac{1}{2}(x^i + iy^i)$ and $\bar{z}^j$, then

$$\omega = \frac{i}{2} \sum_{i,j} h_{ij} z^i \wedge \bar{z}^j.$$

Here

$$h_{ij} = -ih(x_i, y_j).$$

3.2 Differential calculus on manifolds

Lecture 12: Complex calculus

Tue 24 Mar 2026 12:31

Definition 3.2.1. An *almost complex manifold* (X, I) is a smooth manifold X and an $\mathbb{R}$ -linear endomorphism $J : TX \rightarrow TX$ of the tangent bundle with $J^2 = -1$.

Proposition 3.2.2. *Every complex manifold has an almost complex structure.*

Proof. Multiply by i in coordinates. ■

Remark 3.2.3. Not all even dimensional smooth manifolds admit almost complex structures. For example, the standard smooth structure on S^4 does not.

Definition 3.2.4. Let X be an almost complex manifold. Then $T_{\mathbb{C}}X$ denotes the complexification of TX , defined by $T_{\mathbb{C}}X := TX \otimes_{\mathbb{R}} \mathbb{C}$. A priori, this is just a complex vector bundle without a holomorphic structure, even if X is a complex manifold.

Proposition 3.2.5. *Let (X, J) be an almost complex manifold. Then there is a decomposition*

$$T_{\mathbb{C}}X \cong T^{1,0}X \oplus T^{0,1}X$$

of complex vector bundles on X , where the $\mathbb{C}$ -linear extension of J acts by multiplication by i on $T^{1,0}X$ and by multiplication by $-i$ on $T^{0,1}X$.

Proof. One defines

$$T^{1,0}X := \text{Ker}(J - i1 : T_{\mathbb{C}}X \rightarrow T_{\mathbb{C}}X),$$

and similarly for $T^{0,1}X$. One simply decomposes the bundles fiberwise and applies lemma 3.1.8. ■

Definition 3.2.6. Let (X, J) be an almost complex manifold. Then $T^{1,0}X$ and $T^{0,1}X$ are called the *holomorphic* and *antiholomorphic* tangent bundles of X , respectively.

Remark 3.2.7. Let $E \rightarrow X$ be a complex vector bundle with cocycles $\{\psi_{ij}\}$. Then one defines the complex vector bundle $\bar{E} \rightarrow X$ by taking the cocycles to be $\{\bar{\psi}_{ij}\}$. The fibers are naturally isomorphic, but multiplication differs by a sign. Even if E is a holomorphic vector bundle, $\bar{E}$ need not be. Note that $T^{0,1}X \cong \overline{T^{1,0}X}$.

Intuition 3.2.8. We work fiberwise. Let X a complex manifold. We complexify all the tangent spaces $T_p^{\mathbb{C}}X$, and decompose this into holomorphic and antiholomorphic components $T^{1,0}X \oplus T^{0,1}X$. Choose holomorphic coordinates $z_i = x_i + iy_i$ for X . The real tangent space has a basis $\{\partial_{x_i}, \partial_{y_i}\}$, and it makes sense to have the almost complex structure act on these by $J\partial_{x_i} = \partial_{y_i}$. By ?? 3.1.9, the projection $\pi^{1,0}(\partial_{x_i})$ looks like

$$\frac{1}{2} \left(\frac{\partial}{\partial x_i} - iJ \left(\frac{\partial}{\partial x_i} \right) \right) = \frac{1}{2} \left(\frac{\partial}{\partial x_i} - i \frac{\partial}{\partial y_i} \right) = \frac{\partial}{\partial z_i}.$$

This of course means that $dz_i \in (T^{1,0}X)^*$ and $d\bar{z}_i \in (T^{0,1}X)^*$.

Definition 3.2.9. Let (X, J) be an almost complex manifold. Then we define the vector bundles

$$\Lambda_{\mathbb{C}}^k X := \Lambda^k(T_{\mathbb{C}}X)^* \text{ and } \Lambda^{p,q} X := \Lambda^p(T^{1,0}X)^* \otimes_{\mathbb{C}} \Lambda^q(T^{0,1}X)^*.$$

The sheaves of sections are denoted by $\mathcal{A}_X^k$ and $\mathcal{A}_X^{p,q}$, respectively. Global sections of $\mathcal{A}_X^{p,q}$ are called *forms of type (p, q)* . The projections $\mathcal{A}^{\bullet}(X) \rightarrow \mathcal{A}^k(X)$ and $\mathcal{A}^{\bullet}(X) \rightarrow \mathcal{A}^{p,q}(X)$ are denoted by Π^k and $\Pi^{p,q}$, respectively.

Corollary 3.2.10. *We have*

$$\Lambda_{\mathbb{C}}^k X \cong \bigoplus_{p+q=k} \Lambda^{p,q} X, \quad \text{and} \quad \mathcal{A}_X^k \cong \bigoplus_{p+q=k} \mathcal{A}_X^{p,q}.$$

Proof. Immediate by ?? 3.1.13, and pointwise computation of universals in functor categories and for vector bundles. $\blacksquare$

Definition 3.2.11. Let (X, J) be an almost complex manifold. The *exterior derivative* $d : \mathcal{A}_X^k \rightarrow \mathcal{A}_X^{k+1}$ is the $\mathbb{C}$ -linear extension of $d : \Omega^k(X) \rightarrow \Omega^{k+1}(X)$.

Discussion 3.2.12. The exterior derivative is not quite the natural notion of a derivative. Recall that if $\omega = \omega_p dz_{i_1} \wedge \cdots \wedge dz_{i_k}$ is a form, then since d acts by the *real derivatives* x_i ,

$$d\omega = \sum \frac{\partial \omega_p}{\partial x_i} dx_i \wedge dz_{i_1} \wedge \cdots \wedge dz_{i_k}.$$

It is for this reason that $d\omega$ falls in $\mathcal{A}_X^{k+1}$, but we cannot say much about its type: it is a sum of many types. It would be more natural to have a derivative that acts by taking partial derivatives of ω_p with respect to z_i or $\bar{z}_i$. We would think of the former as going $\mathcal{A}_X^{p,q} \rightarrow \mathcal{A}_X^{p+1,q}$, and similarly for the latter.

Definition 3.2.13. Let (X, J) be an almost complex manifold. For all p, q , define

$$\partial := \Pi^{p+1,q} \circ d : \mathcal{A}_X^{p,q} \rightarrow \mathcal{A}_X^{p+1,q}$$

$$\bar{\partial} := \Pi^{p,q+1} \circ d : \mathcal{A}_X^{p,q} \rightarrow \mathcal{A}_X^{p,q+1}.$$

Discussion 3.2.14. Let $f : X \rightarrow \mathbb{C}$ be a smooth function on a complex manifold X . Let $z_i = x_i + iy_i$ be coordinates. Then in these coordinates,

$$df = \sum_i \frac{\partial f}{\partial x_i} dx_i + \sum_i \frac{\partial f}{\partial y_i} dy_i.$$

An annoying but not unexpected calculation shows that

$$df = \sum_i \frac{\partial f}{\partial x_i} dx_i + \sum_i \frac{\partial f}{\partial y_i} dy_i = \sum_i \frac{\partial f}{\partial z_i} dz_i + \sum_i \frac{\partial f}{\partial \bar{z}_i} d\bar{z}_i.$$

This is easily checked for functions in one complex variable, which illustrates how the statement follows in general. As noted in ?? 3.2.8, we have $dz_i \in (T^{1,0}X)^*$ and $d\bar{z}_i \in (T^{0,1}X)^*$, so the projections map

$$\partial f = \sum_i \frac{\partial f}{\partial z_i} dz_i$$

$$\bar{\partial} f = \sum_i \frac{\partial f}{\partial \bar{z}_i} d\bar{z}_i.$$

Hence f is holomorphic if and only if $\bar{\partial} f = 0$.

More generally, let $\omega = f dz_{i_1} \wedge \cdots \wedge dz_{i_p} \wedge d\bar{z}_{j_1} \wedge \cdots \wedge d\bar{z}_{j_q}$ be a form of type (p, q) . Then since d is a derivation and $d^2 = 0$, we have

$$\begin{aligned} d\omega &= df \wedge dz_{i_1} \wedge \cdots \wedge d\bar{z}_{j_q} \\ &= \sum_i \frac{\partial f}{\partial x_i} dx_i \wedge dz_{i_1} \wedge \cdots \wedge d\bar{z}_{j_q} \\ &= \left(\sum_i \frac{\partial f}{\partial z_i} dz_i + \sum_i \frac{\partial f}{\partial \bar{z}_i} d\bar{z}_i \right) dz_{i_1} \wedge \cdots \wedge d\bar{z}_{j_q} \end{aligned}$$

Hence

$$\begin{aligned} \partial\omega &= \sum_i \frac{\partial f}{\partial z_i} dz_i \wedge dz_{i_1} \wedge \cdots \wedge d\bar{z}_{j_q} \\ \bar{\partial}\omega &= \sum_i \frac{\partial f}{\partial \bar{z}_i} d\bar{z}_i \wedge dz_{i_1} \wedge \cdots \wedge d\bar{z}_{j_q}. \end{aligned}$$

Additionally, we have $d = \partial + \bar{\partial}$.

Lemma 3.2.15. *Let X be a complex manifold (NOT ALMOST COMPLEX!). Then*

1. $d = \partial + \bar{\partial}$;
2. $\partial^2 = \bar{\partial}^2 = 0$ and $\partial\bar{\partial} = -\bar{\partial}\partial$;
3. They satisfy the graded Leibniz rule

$$\partial(\alpha \wedge \beta) = \partial\alpha \wedge \beta + (-1)^{|\alpha|} \alpha \wedge \partial\beta$$

$$\bar{\partial}(\alpha \wedge \beta) = \bar{\partial}\alpha \wedge \beta + (-1)^{|\alpha|} \alpha \wedge \bar{\partial}\beta$$

for $\alpha, \beta \in \mathcal{A}_X^\bullet$.

Proof. (1) Done in ??.

(2) We use $d^2 = 0$ to obtain

$$d^2 = (\partial + \bar{\partial}) \circ (\partial + \bar{\partial}) = \partial^2 + \partial\bar{\partial} + \bar{\partial}\partial + \bar{\partial}^2.$$

It suffices to show that $\partial^2 = \bar{\partial}^2 = 0$. Indeed, let $\omega = f dz_{i_1} \wedge \cdots \wedge d\bar{z}_{j_q}$. Let η be $dz_{i_1} \wedge \cdots \wedge d\bar{z}_{j_q}$ such that $\omega = f \wedge \eta$. Note that $d\eta = 0$, and thus $\partial\eta = 0$. Then using (3), we have

$$\partial^2\omega = \partial(\partial f \wedge \eta) = \partial^2 f \wedge \eta.$$

The statement is then analogous to the proof that $d^2 f = 0$.

(3) The exterior derivative satisfies it. It follows immediately. ■

Proposition 3.2.16. *Let $f : X \rightarrow Y$ be a holomorphic map between complex manifolds. Then the pullback induces a $\mathbb{C}$ -linear map $f^* : \mathcal{A}^{p,q}(Y) \rightarrow \mathcal{A}^{p,q}(X)$ which commutes with ∂ and $\bar{\partial}$.*

Proof. The pullback exists by smooth geometry and satisfies $d \circ f = f \circ d$. If it's holomorphic, it commutes with the projections. ■

This proposition identifies $\Omega^\bullet(X) \cong \Lambda^{\bullet,0}(X)$ as complex vector bundles, and thus holomorphic sections of $\Omega^\bullet(X)$ are sections of $\Lambda^{\bullet,0}X$.

Proposition 3.2.17. *Let X be a complex manifold. Then the space of holomorphic p -forms on X is the subspace of $\alpha \in \mathcal{A}^{p,0}(X)$ such that $\bar{\partial}\alpha = 0$.*

Definition 3.2.18. Let X be a complex manifold. Then the (p, q) -Dolbeault cohomology is the $\mathbb{R}$ -vector space

$$H^{p,q}(X) := \frac{\text{Ker}(\bar{\partial} : \mathcal{A}^{p,q}(X) \rightarrow \mathcal{A}^{p,q+1}(X))}{\text{Im}(\bar{\partial} : \mathcal{A}^{p,q-1}(X) \rightarrow \mathcal{A}^{p,q}(X))}.$$

Lecture 13: Kähler metrics

Tue 24 Mar 2026 12:31

Given $U \subseteq \mathbb{C}^n$ and a metric g compatible with the complex structure, we have the fundamental form $\omega = g(J-, -) \in \mathcal{A}_{\mathbb{C}}^{1,1}$. We say g is *Kähler* if $d\omega = 0$. Recall that the Hermitian metric associated to g is $h = g - i\omega$. We say this osculates to the identity of order 2 if for all $p \in U$, there are coordinates centered at p such that the Taylor expansion of h is the identity up to order 2.

Proposition 3.2.19. $d\omega = 0$ iff h osculates to the identity of order 2.

Proof. ($\Leftarrow$) If it osculates then $d\omega = 0$.

($\Rightarrow$) Conversely assume $d\omega = 0$. Write $\omega = \frac{i}{2}h_{ij}dz_i \wedge d\bar{z}_j$. At the origin, h_{ij} is Hermitian, and after a linear change in coordinates we may thus assume $h_{ij}(0) = I$. We Taylor expand to order n :

$$h_{ij} = \delta_{ij} + \sum_{k=1}^n a_{ijk}z_k + \sum_{k=1}^n a'_{ijk}\bar{z}_k + O(|z|^2).$$

These have

$$a_{ijk} = \frac{\partial h_{ij}}{\partial z_k}(0), \quad a'_{ijk} = \frac{\partial h_{ij}}{\partial \bar{z}_k}(0).$$

We have

$$\begin{aligned} d\omega &= \frac{i}{2}d(h_{ij})dz_i \wedge d\bar{z}_j \\ &= \frac{i}{2} \left(\sum_{k=1}^n a_{ijk}z_k dz_k + \sum_{k=1}^n a'_{ijk}d\bar{z}_k \right) \wedge dz_j \wedge d\bar{z}_j + O(|z|^2). \end{aligned}$$

The terms in the sum have types $(2, 1)$ and $(1, 2)$, so they have to cancel. We use $a_{ijk} = a_{kji}$ and $a'_{ijk} = a'_{ikj}$. Also $h_{ij} = \overline{h_{ji}}$ because it's Hermitian at 0.

Introduce coordinates $\omega_j = z_j + \sum_{i,k=1}^n a_{ijk}z_i z_k$. An unpleasant calculation shows that ω in the new coordinates is of the form

$$\omega = \frac{i}{2} \sum_{j=1}^n d\omega_j \wedge d\bar{\omega}_j + O(|\omega|^2).$$

■

This yields geometric meaning for the condition that a manifold be Kähler.

For complex manifolds, $d = \partial + \bar{\partial}$. This is not true for almost complex manifolds. However, we have conditions on when it holds.

Theorem 3.2.20 (Newlander-Nierenberg). *Let (X, J) be an almost complex manifold. The following are equivalent.*

1. X has a complex manifold structure giving rise to J .
2. $d = \partial + \bar{\partial}$ on $\mathcal{A}^{1,0}$;
3. $d = \partial + \bar{\partial}$ on $\mathcal{A}^\bullet$;
4. $\bar{\partial}^2 = 0$;
5. $[T_X^{1,0}, T_X^{1,0}] \subseteq T_X^{1,0}$.

A highly nontrivial and highly analytic proof exists which we do not provide.

Chapter 4

Sheaf cohomology

Lecture 14: Sheaves

Thu 26 Mar 2026 12:31

4.1 Sheaf cohomology

Let $\mathcal{F}$ be a sheaf on a space X valued in a concrete abelian category. Let $U \subseteq X$ be open. Then there is a map to the product of stalks

$$\mathcal{F}(U) \rightarrow \prod_{x \in U} \mathcal{F}_x$$

given by $s \mapsto \{s_x\}_{x \in U}$, where s_x is the germ of s at x . This map is injective since its kernel is trivial: if $s_x = 0$, then there is $V \subseteq U$ containing x such that $s|_V = 0$; the gluing axiom yields $s = 0$.

Notation 4.1.1. We write the category of presheaves on a space X valued in $\mathcal{C}$ as $\mathcal{Pshf}_X(\mathcal{C})$. We often leave the $\mathcal{C}$ implicit and write $\mathcal{Pshf}_X$. Similarly, we write $\mathcal{Shf}_X(\mathcal{C}) = \mathcal{Shf}_X$ for the full subcategory of $\mathcal{Pshf}_X(\mathcal{C})$ spanned by sheaves.

Construction 4.1.2. Let $\mathcal{F}$ be a presheaf on a space X . Let $U : \mathcal{Shf}_X \rightarrow \mathcal{Pshf}_X$ be the forgetful functors. Then the *sheafification* $\mathcal{F}^+$ of $\mathcal{F}$ is the sheaf which is universal in the sense that there is a natural transformation $i : \mathcal{F} \Rightarrow \mathcal{F}^+$ whose precomposition induces isomorphisms

$$i^* : \mathcal{Shf}_X(\mathcal{F}^+, \mathcal{G}) \cong \mathcal{Pshf}_X(\mathcal{F}, U(\mathcal{G}))$$

for all sheaves $\mathcal{G}$, natural in $\mathcal{F}$ and $\mathcal{G}$. More precisely, $-^+ \dashv U$, meaning the sheafification of a sheaf is the free sheaf on a presheaf.

A concrete discussion is as follows. Let $U \subseteq X$ be open, and define $F^+(U)$ as the set of all functions $t : U \rightarrow \prod_{x \in U} \mathcal{F}_x$ having the property that for all $x \in U$, there exists $V \ni x$ and a section $s \in \mathcal{F}(U)$ such that $t(x) = s_x$. This is called the *Étalé space*.

Define the set

$$E := \prod_{x \in X} \mathcal{F}_x.$$

Topologize it as follows. If $s \in \mathcal{F}(U)$, then define the map $\tilde{s} : U \rightarrow E$ by $\tilde{s}(x) = s_x$. The topology on E is generated by taking the following as a basis:

$$\mathcal{B} := \{\tilde{s}(U) \mid s \in \mathcal{F}(U), U \in \mathcal{O}pen(X)\}.$$

The projection $\pi : E \rightarrow X$ is then a local homeomorphism, and since it is surjective, we can consider sections $\Gamma(U, E)$ of an open set $U \subseteq X$. The sheafification $\mathcal{F}^+$ can be defined as the sheaf of sections

$$\mathcal{F}^+ \cong \Gamma(-, E).$$

Remark 4.1.3. Sheaves are stable under *most* relevant (co)limits such that pointwise construction of (co)limits yields an abelian category structure for $\text{Shf}_X(\mathcal{A})$ whenever $\mathcal{A}$ is abelian. For example, if $\phi : \mathcal{F} \rightarrow \mathcal{G}$ is a natural transformation, then

$$\text{Ker } \phi(U) := \text{Ker}(\phi_U : \mathcal{F}(U) \rightarrow \mathcal{G}(U))$$

is a sheaf. To show this, let $\{U_i\}_{i \in I}$ an open cover of U , let $\{s_i\}_{i \in I}$ with $s_i \in \text{Ker } \phi(U_i) = \text{Ker } \phi_{U_i}$. Assume that $s_i|_{U_i \cap U_j} = s_j|_{U_i \cap U_j}$ for all i, j . Then we identify $\text{Ker } \phi_{U_i} \subseteq \mathcal{F}(U_i)$ to obtain that $s_i|_{U_i \cap U_j} = s_j|_{U_i \cap U_j}$ as elements of $\mathcal{F}$, since the restrictions agree in the kernel. Thus there exists $s \in \mathcal{F}(U)$ with $s|_{U_i} = s_i$. We show $\phi_U(s) = 0$. Indeed, by naturality of ϕ we obtain

$$\phi_U(s)|_{U_i} = \phi_U(s_i) = 0.$$

Thus $\phi_U(s)$ is given by gluing together $0 \in \mathcal{G}(U_i)$ along the open cover $\{U_i\}_{i \in I}$, locality yields $\phi_U(s) = 0$.

However, Shf_X is not stable under all colimits. For example, $\text{Im } \phi$ is not a sheaf. Since $\text{Im} = \text{Ker coker}$, this tells us that the cokernel need not be a sheaf. We instead define the image $\text{Im } \phi$ to be the sheafification of the pointwise image of ϕ .

More generally, we note the following: if $\mathcal{F}$ is a presheaf, then it is a sheaf if and only if for all opens $U_a, U_b \subseteq X$,

$$\mathcal{F}(U) \cong \mathcal{F}(U_a) \times_{\mathcal{F}(U_a \cap U_b)} \mathcal{F}(U_b).$$

Thus Shf_X is stable under limits since limits commute, but not necessarily under colimits. Thus we by convention re-define colimits in Shf_X by taking the sheafification of the relevant colimit in Pshf . Since sheafification is left-adjoint, it preserves colimits, and thus is indeed a colimit.

Definition 4.1.4. Since the image of a morphism of sheaves (as a categorical definition) is *not* the image as a morphism of presheaves, we define a morphism $\phi : \mathcal{F} \rightarrow \mathcal{G}$ of sheaves to be *surjective/epi* if $\text{Im } \phi \cong \mathcal{G}$.

Exercise 4.1.5. With notation as in definition 4.1.4, the following are equivalent.

1. ϕ is surjective.
2. $\phi_x : \mathcal{F}_x \rightarrow \mathcal{G}_x$ is surjective as a function for all $x \in X$.
3. For all open $U \subseteq X$ and $s \in \mathcal{G}(U)$, there exists $x \in U$, an open set $x \in V \subseteq U$, and a section $t \in \mathcal{F}(V)$ such that $\phi(t) = s|_V$. In particular, locally a surjective morphism of sheaves is surjective in the usual sense.

Example 4.1.6. Consider the sheaf $\mathcal{O}_X^*$ of nowhere vanishing holomorphic functions on a complex manifold X . This takes values in multiplicative abelian groups, so we have a natural transformation $\exp : \mathcal{O}_X \rightarrow \mathcal{O}_X^*$ which maps $f : U \subseteq X \rightarrow \mathbb{C}$ to $\exp(2\pi i f) : U \rightarrow \mathbb{C}$. This map is very important. We note that f is in the kernel of $\exp_U$ if and only if $f(z) = n$ for some $n \in \mathbb{Z}$ and all $z \in U$. Hence $\text{Ker } \exp \cong \mathbb{Z}_X$ is the constant sheaf on X .

We can also see that $\exp$ is surjective. More precisely, for all $x \in X$ and $g \in \mathcal{O}_X^*$ defined near x , we can find some $f \in \mathcal{O}_X^*$ defined in a neighborhood of x with $\exp(2\pi i f) = g$. Hence we are claiming that locally a logarithm can be defined.

To see this, let $g(x) \in \mathbb{C}$. Since this is nonzero, there is a neighborhood which does not contain 0, and thus sufficiently near as to avoid a branch cut. Thus $\ln g$ can be defined on the preimage $g^{-1}(U)$ of some sufficiently small open neighborhood $U \ni g(x)$.

It follows that

$$0 \longrightarrow \mathbb{Z}_X \longrightarrow \mathcal{O}_X \xrightarrow{\exp} \mathcal{O}_X^* \longrightarrow 0$$

is an exact sequence of sheaves.

Example 4.1.7. Let X be a complex manifold. Recall that $\mathcal{A}_X^{p,q}$ is the sheaf of smooth (p, q) -forms on X . Write $\mathcal{A}^{p,\bullet}$ for the complex

$$0 \longrightarrow \mathcal{A}_X^{p,0} \xrightarrow{\bar{\partial}} \mathcal{A}_X^{p,1} \xrightarrow{\bar{\partial}} \mathcal{A}_X^{p,2} \xrightarrow{\bar{\partial}} \dots$$

As seen before, this is a complex since $\bar{\partial}^2 = 0$. We claim this sequence is exact at $\mathcal{A}^{p,q}$ for all $q \geq 1$. This follows from the $\bar{\partial}$ -Poincaré lemma, which says that on a polydisk, every $\bar{\partial}$ -closed (p, q) form with $q > 0$ is $\bar{\partial}$ -exact.

Definition 4.1.8. A sheaf $\mathcal{F}$ on a complex manifold is an $\mathcal{O}_X$ -module if $\mathcal{F}(U)$ is an $\mathcal{O}_X(U)$ -module for all open sets U . More generally, if $(X, \mathcal{O}_X)$ is a locally ringed space (meaning $\mathcal{O}_X$ is a sheaf of rings, called the structure sheaf, and usually some kind of sheaf of functions), then an $\mathcal{O}_X$ -module is a sheaf $\mathcal{F}$ on X such that $\mathcal{F}(U)$ is an $\mathcal{O}_X(U)$ -module for all open sets $U \subseteq X$.

Example 4.1.9. For example, if $\pi : E \rightarrow X$ is a holomorphic vector bundle, then the sheaf of sections $\mathcal{E} = \Gamma(-, E)$ is an $\mathcal{O}_X$ -module (since we can just multiply two holomorphic functions).

Example 4.1.10. Let $x \in X$. Define the *skyscraper sheaf at x* as the sheaf $S_x(U)$ to be $\mathbb{C}$ if $x \in U$ and 0 otherwise.

Consider a category of sheaves $\text{Shf}_X(\mathcal{A})$ on an abelian category. The global sections functor $\mathcal{F} \mapsto \Gamma(X, \mathcal{F})$ is not exact (it is right adjoint to the constant sheaf functor). More concretely, surjectivity at the level of sheaves and at the level of sets of sections are *different*, meaning we only get that $\Gamma(X, -)$ preserves kernels, but not surjectivity. Thus it has right derived functors.

Definition 4.1.11. Let $\mathcal{F}$ be a sheaf on X valued in an abelian category $\mathcal{A}$. The *sheaf cohomology* is given by the right derived functors of the global sections functor

$$H^\bullet(X, \mathcal{F}) := R^n \Gamma(X, \mathcal{F}).$$

Lecture 15: Examples of sheaf cohomology

Tue 31 Mar 2026 12:25

To compute the right derived functors $R^n \Gamma$, we can resolve $\mathcal{F}$ by injective sheaves. Unfortunately, injective sheaves suck. Luckily, we can more generally find an *acyclic resolution*. Let $F : \mathcal{A} \rightarrow \mathcal{B}$ be a left exact functor of abelian categories. Then some object $x \in \mathcal{A}$ is *acyclic for F* if $R^n F(x) = 0$ for all $n > 0$.

Fact: for sheaves, sheaves of smooth sections of vector bundles are $\Gamma(X, -)$ -acyclic. Intuitively, this is because locally these sheaves look like direct sums of the structure sheaf for smooth manifolds C^∞ , and thus admit partitions of unity, allowing global to local information transfer, deleting the higher homological information. Thus if we can resolve a sheaf by sheaves of this type, we can compute the cohomology by applying $\Gamma(X, -)$ to this resolution.

Example 4.1.12. Let M a smooth n -manifold. Then we get an exact sequence of sheaves

$$0 \longrightarrow \Omega_M^0 \xrightarrow{d} \Omega_M^1 \xrightarrow{d} \cdots \xrightarrow{d} \Omega_M^n \xrightarrow{0} 0.$$

This is exact by the Poincare lemma in all nonzero degrees: any form that is closed is locally exact. Hence kernels are locally images, and by ?? 4.1.5 images are local. However, it is not exact in the lowest degree, since it has kernel $\mathbb{R}$: constant functions. Thus the complex is quasi-isomorphic to the (sheafification of the) constant sheaf $\mathbb{R}$, and is thus a resolution. This yields

$$H_{\text{shf}}^n(M, \mathbb{R}) = R^n \Gamma(M, \mathbb{R}) \cong H^n(\Omega_M^\bullet(M)).$$

The de Rham cohomology computes the cohomology of the constant sheaf $\mathbb{R}$. Of course, this is just $H_{\text{Sing}}^n(M, \mathbb{R})$. This explains why the de Rham complex is topological in nature: it is computing the cohomology of something with no local structure.

Example 4.1.13. Recall from example 4.1.7 that if X is a complex n -manifold, then we have the Dolbeault complex for global sections

$$0 \longrightarrow \mathcal{A}_X^{p,0}(X) \xrightarrow{\bar{\partial}} \mathcal{A}_X^{p,1}(X) \xrightarrow{\bar{\partial}} \cdots \xrightarrow{\bar{\partial}} \mathcal{A}_X^{p,n}(X) \xrightarrow{\bar{\partial}} 0,$$

which is exact at q for all $q \geq 1$, since once again the $\bar{\partial}$ -Poincare lemma says any $\bar{\partial}$ -closed $(p, q \geq 1)$ -form is locally $\bar{\partial}$ -exact. However, in degree 0, $\bar{\partial}$ has a kernel given by holomorphic p -forms (proposition 3.2.17). Thus this is a resolution of the sheaf of holomorphic p -forms, denoted Ω_X^p , and we obtain that the (p, q) -Dolbeault cohomology is the cohomology of this sheaf. Therefore

$$H_{\text{shf}}^\bullet(X, \Omega_X^p) \cong H^\bullet(\mathcal{A}_X^{p,\bullet}(X)).$$

Thus the sheaf cohomology of Ω_X^p is the Dolbeault cohomology $(p, \bullet)$. This is not a homotopy invariant, unlike de Rham cohomology. To see this, consider for example $H^\bullet(\mathbb{C}^n, \mathcal{O})$. There are many global holomorphic functions $\mathbb{C}^n \rightarrow \mathbb{C}$, but the homotopy type of $\mathbb{C}^n$ is a point, and there are only constant holomorphic functions $* \rightarrow \mathbb{C}$.

Remark 4.1.14. There is a generalization of Dolbeault cohomology. Let $\pi : E \rightarrow X$ a holomorphic vector bundle. We can consider the sheaf $\mathcal{A}^{p,q}(E)$ of (p, q) -forms valued in E . By this, we mean elements of the form $\sum \alpha \otimes e$, where $\alpha \in \mathcal{A}^{p,q}$ and e is a local holomorphic section of E . Hence $\mathcal{A}^{p,q} \otimes_{\mathcal{O}_X} \mathcal{E} \cong \mathcal{A}^{p,q}(E)$.

Present E via the cocycle $\{g_{ij}\}$ wrt some open cover $\{U_i\}$. Sections of $\mathcal{A}^{p,q}(E)$ are $\text{rk}(E)$ -vectors of (p, q) -forms that transform according to the cocycle, i.e. for all $i \in I$, there is $s_i \in \Gamma(U_i, (\mathcal{A}^{p,q})^{\oplus r})$ and $s_i = g_{ij}s_j$. Then $\bar{\partial}$ is defined on the sheaf:

$$\bar{\partial}s_i = \bar{\partial}(g_{ij}s_j) = \bar{\partial}g_{ij} \cdot s_j + g_{ij}\bar{\partial}s_j = g_{ij}\bar{\partial}s_j,$$

since g_{ij} is holomorphic. Thus we have the required relation for $\bar{\partial}s$ to be a section. By the $\bar{\partial}$ -Poincare lemma, we obtain the same exactness relationship. The kernel of the bottom degree differential is just holomorphic sections of $E \otimes_{\mathbb{C}} \mathcal{O}_{\text{hol}}^p$, with sheaf denoted $\mathcal{E} \otimes \Omega_{\text{hol}}^p$. Therefore

$$H_{\text{shf}}^\bullet(X, \mathcal{E} \otimes \Omega_{\text{hol}}^p) \cong H^\bullet(\mathcal{A}^{p,\bullet}(E), \bar{\partial}).$$

This feels wrong

In general, $\mathcal{A}^{p,q}(E)$ is ∞ -dimensional, and difficult to actually find the cohomology of. We need better tools for computing cohomology. In de Rham cohomology, we use homotopy invariance to actually do computations by relating it to singular cohomology. Sheaves have another cohomology theory that often coincides with the one we have been using, known as Čech cohomology.

4.2 Čech cohomology

Lecture 16: Čech cohomology

Tue 07 Apr 2026 12:34

To motivate Čech cohomology, let X a topological space and $\mathcal{F}$ a sheaf on X . Consider an open cover $\mathcal{U} := \{U_i\}_{i \in I}$ by a linearly ordered index set I . We will try to compute global sections of $\mathcal{F}$ using the open cover $\mathcal{U}$. Using the gluing property, if we have a collection $\{s_i \in \mathcal{F}(U_i)\}_{i \in I}$ of sections which agree on pairwise overlaps $s_i|_{U_i \cap U_j} = s_j|_{U_i \cap U_j}$, then there is a unique $s \in \mathcal{F}(X)$ such that $s|_{U_i} = s_i$. Thus we can glue together compatible local sections along the open cover.

We want to write down the set of all global sections by writing down the sets of all compatible local sections along $\mathcal{U}$. We want to do this in more homological terms. Define the zeroth cochains $\check{C}^0(\mathcal{U}, \mathcal{F})$ to be

$$\prod_{i \in I} \mathcal{F}(U_i).$$

This is a first approximation to global sections, but it ignores compatibility concerns. We write

$$U_{i_0 \dots i_n} := U_{i_0} \cap \dots \cap U_{i_n}.$$

Then the first cochains are

$$\check{C}^1(\mathcal{U}, \mathcal{F}) := \prod_{i_0 < i_1} \mathcal{F}(U_{i_0 i_1}).$$

We want to define a differential δ such that the kernel $\text{Ker } \delta$ consists of sections which agree on the overlaps. Hence we want $\text{Ker } \delta = \Gamma(X, \mathcal{F}) = H^0(X, \mathcal{F})$. Thus we define

$$(\delta\gamma)_{ij} = \gamma_i - \gamma_j.$$

Here $\gamma = \{\gamma_i \in \mathcal{F}(U_i)\}_{i \in I}$, and we write $\gamma_{ij} = \gamma|_{U_{ij}}$. We can extend this to higher degrees:

$$\check{C}^p(\mathcal{U}, \mathcal{F}) := \prod_{i_0 < \dots < i_p} \mathcal{F}(U_{i_0 \dots i_p})$$

with differential $\delta : \check{C}^p(\mathcal{U}, \mathcal{F}) \rightarrow \check{C}^{p+1}(\mathcal{U}, \mathcal{F})$ given by

$$(\delta\gamma)_{i_0 \dots i_{p+1}} = \sum_{j=0}^p (-1)^j \gamma_{i_0 \dots \widehat{i_j} \dots i_{p+1}}.$$

This yields a cochain complex $\check{C}^\bullet(\mathcal{F}, \mathcal{U})$. Since the 0th homology is $\Gamma(X, \mathcal{F})$, this complex is in some sense *trying* to resolve Γ . However, this complex is not quite exact.

Definition 4.2.1. The complex $(\check{C}^\bullet(\mathcal{U}, \mathcal{F}), \delta)$ is called the *Čech complex* of the sheaf $\mathcal{F}$ with respect to the open cover $\mathcal{U}$.

The choice of open cover is important. Consider $X = \mathbb{P}^1 \cong S^2$, and consider the constant sheaf $\mathcal{F} = \mathbb{C}$. Recall $H_{\text{shf}}^\bullet(X, \mathbb{C})$ is singular cohomology $H_{\text{Sing}}^\bullet(X, \mathbb{C})$. Recall that S^2 has cohomology $\mathbb{C}$ concentrated in degrees 0 and 2. Let 0 denote the north pole and ∞ denote the south pole. Then

$$U_0 := X \setminus \{\infty\} \cong \mathbb{R}^2,$$

$$U_1 := X \setminus \{0\} \cong \mathbb{R}^2,$$

$$U_{01} := X \setminus \{0, \infty\} \cong \mathbb{C} \setminus \{0\}.$$

Since constant sheaves are locally constant, they are constant on connected sets. Thus

$$\mathcal{F}(U_0) \cong \mathcal{F}(U_1) \cong \mathbb{C}.$$

Similarly, $\mathbb{C} \setminus \{0\}$ is connected, so $\mathcal{F}(U_{01}) \cong \mathbb{C}$. The differential $\delta : \mathcal{F}(U_0) \times \mathcal{F}(U_1) \rightarrow \mathcal{F}(U_{01})$ is simply subtraction $(a, b) \mapsto b - a$. Hence δ is surjective, since any complex number can be written as a difference of complex numbers. The kernel is pairs $(a, b) = (a, a)$, hence $\text{Ker } \delta \cong \mathbb{C}$. We find that the cohomology is simply

$$H^0(\check{C}^\bullet(\{U_0, U_1\}, \mathcal{F})) = \mathbb{C},$$

and the cohomology vanishes in other degrees. This is not simplicial cohomology.

What has gone wrong? Clearly, the cover is not fine enough. For example, we need the complex to not vanish in degree 2, meaning we need at least 3 open sets. Another idea is as follows. The complex was sort of trying to resolve $\Gamma(X, -)$. Normally, our resolutions need to be $\Gamma(X, -)$ -acyclic, so we need an analog. In particular, we need the sheaf cohomology of all intersections $H_{\text{shf}}^n(U_{i_0 \dots i_p}, \mathcal{F})$ to vanish for $n > 0$. This fails here: $H_{\text{shf}}^1(\mathbb{C}^*, \mathbb{C})$ is the singular cohomology, which is nonvanishing since $\mathbb{C}^*$ has the homotopy type of S^1 .

Technically, Čech cohomology is defined as the colimit over *all* covers

$$H^\bullet(\check{C}(\mathcal{F})) = \text{colim}_{\mathcal{U}} H^\bullet(\check{C}(\mathcal{U}, \mathcal{F})).$$

No one is going to enumerate all open covers, so we look for sufficiently nice open covers.

Theorem 4.2.2 (Serre). *Let X be a complex manifold. If $\mathcal{F}$ is a coherent analytic sheaf (a sheaf of $\mathcal{O}_X$ -modules) that locally looks like a cokernel*

$$\text{Coker}(\mathcal{O}_X^{\oplus n} \rightarrow \mathcal{O}_X^{\oplus m}).$$

Google coherent sheaves; analytic coherent is coherent in the holomorphic world. Then on a smooth algebraic variety X , any affine open cover suffices.

Theorem 4.2.3 (GAGA). *If X is a projective manifold (embeddable in projective space), the cohomology*

$$H_{\text{shf}}^\bullet(X, \mathcal{O}_X^{\text{alg}}) \cong H_{\text{shf}}^\bullet(X, \mathcal{O}_X),$$

where $\mathcal{O}_X^{\text{alg}}$ is the sheaf of algebraic functions on X . More generally, this holds for coherent sheaves $\mathcal{F}$.

Example 4.2.4. Consider $X = \mathbb{P}^1$. We choose the sheaf $\mathcal{O}_X$ of holomorphic functions, but by GAGA, we can use $\mathcal{O}_X^{\text{alg}}$. We use $U_0 = X \setminus \{0\}$ and $U_1 = X \setminus \{\infty\}$ once again, and $U_{01} = \mathbb{C}^*$ is the circle. This cover is okay to use here because we are now working with algebraic functions, and all three of these sets are affine varieties. There is no higher cohomology on affine varieties, so these open sets pass the acyclicity test for algebraic functions. Choose a coordinate system near 0 such that algebraic functions on U_0 are polynomials $\mathbb{C}[z]$. We then identify algebraic functions on U_1 with polynomials $\mathbb{C}[z^{-1}]$ near ∞ . Algebraic functions on $\mathbb{C}^*$ are Laurant polynomials $\mathbb{C}[z, z^{-1}]$.

The differential δ maps polynomials as

$$(p(z), q(z^{-1})) \mapsto q(z^{-1}) - p(z).$$

This map is surjective *because we are working in polynomials*. The kernel is once again $\mathbb{C}$, viewed as pairs of constant functions $a \mapsto (a, a)$. Thus $H_{\text{shf}}^0(X, \mathcal{O}_X^{\text{alg}})$ is $\mathbb{C}$. The cohomology vanishes in $n > 0$.

Definition 4.2.5. Let X be a topological space. An open cover $\{U_i\}_{i \in I}$ is *good* is a cover when all open sets U_i and all finite n -fold intersections thereof are contractible.

Lecture 17: Čech cohomology, continued

Thu 09 Apr 2026 12:30

Theorem 4.2.6 (Čech). *If $H^n(\{U_{i_0}\}, \mathcal{F}) = 0$ for $n > 0$ for all $U_{i_j} \in \mathcal{U}$, then*

$$H^\bullet(\check{C}(\mathcal{U}, \mathcal{F})) = H_{\text{shf}}^\bullet(X, \mathcal{F}).$$

This can always be done on manifolds. For example, give a manifold a Riemannian metric, and take geodesic balls.

Let X be a complex manifold. We consider the relation $H^1(X, \mathcal{O}_X^*)$ and the Picard group $\text{Pic}(X)$, which is

$\text{Pic}(X) = \text{Group of isomorphism classes of holomorphic line bundles on } X.$

The group operation is the tensor product. We generally abuse notation and identify a line bundle with its sheaf of sections. The inverse is given by taking the dual: $\mathcal{L} \otimes_{\mathcal{O}_X} \mathcal{L}^* \cong \text{End}_{\mathcal{O}_X}(\mathcal{L})$ has a canonical global section given by the identity section $s_x = 1_{\mathcal{L}}$. This gives a global, nowhere vanishing frame via a single section, which yields a global trivialization of $\mathcal{L} \otimes_{\mathcal{O}_X} \mathcal{L}^* \cong X \times \mathbb{R}$. It is simply the trivial bundle.

We sketch why $H_{\text{shf}}^1(X, \mathcal{O}_X^*) \cong \text{Pic}(X)$. Choose a cover $\mathcal{U} = \{U_i\}_{i \in I}$ which trivializes a line bundle $\mathcal{L}$; i.e. $\mathcal{L}|_{U_i} \cong \mathcal{O}_X$. Trivializations of $\mathcal{L}$ on U_i correspond to holomorphic frams $s_i \in \Gamma(U_i, \mathcal{L})$. Based on this, we can define the cocycle $g_{ij} = \frac{s_i}{s_j}$, where we are defining this fraction as the proportionality of the function defining s_i and s_j on $U_i \cap U_j$.

Since $\mathcal{O}_X^*$ is a multiplicative group, the differential $(\delta g)_{ijk}$ of a cocycle is given by

$$(\delta g)_{ijk} = g_{jk}g_{ik}^{-1}g_{ij} = 1,$$

since $g_{ij}g_{jk} = g_{ik}$. Thus g_{ij} is a cocycle in the Čech cohomology $\check{H}^1(\mathcal{U}, \mathcal{O}_X^*)$. We show that this is invariant of our choice of frame. When we switch frames, we multiply s_i by a nonvanishing function. Then there is something and it works. The point is that it descends to a well-defined cohomology class. A converse can be shown using some facts of colimits, so we get that H^1 is in correspondence with cocycles, which is in correspondence with nonvanishing holomorphic functions, since that is what a cocycle (of a line bundle) is.

Consider the long exact sequence for the global sections functor, which we write as Γ_X : if

$$0 \longrightarrow \mathcal{F} \longrightarrow \mathcal{G} \longrightarrow \mathcal{H}$$

is exact, then there is an exact triangle

$$\begin{array}{ccc} H_{\text{shf}}^\bullet(X, \mathcal{F}) & \longrightarrow & H_{\text{shf}}^\bullet(X, \mathcal{G}) \\ \uparrow +1 & \swarrow & \\ H_{\text{shf}}^\bullet(X, \mathcal{H}) & & \end{array}$$

In particular, there is a long exact sequence

$$\begin{array}{ccccccc} 0 & \longrightarrow & H_{\text{shf}}^0(X, \mathbb{Z}_X) & \longrightarrow & H_{\text{shf}}^0(X, \mathcal{O}_X) & \longrightarrow & H_{\text{shf}}^0(X, \mathcal{O}_X^*) \\ & & & & & \searrow & \\ & & & & & & H_{\text{shf}}^1(X, \mathbb{Z}_X) \longleftarrow H_{\text{shf}}^1(X, \mathcal{O}_X) \longrightarrow H_{\text{shf}}^1(X, \mathcal{O}_X^*) \longrightarrow \cdots \end{array}$$

We apply this to $X = \mathbb{P}^1 \cong S^2$. Since it is connected, since $H_{\text{shf}}^\bullet(X, \mathbb{Z}_X) \cong H_{\text{Sing}}^\bullet(X, \mathbb{Z})$, we know $H_{\text{shf}}^\bullet(X, \mathbb{Z}_X)$. We recall by ?? that the cohomology $H_{\text{shf}}^\bullet(\mathbb{P}^1, \mathcal{O}_X)$ is $\mathbb{C}$ concentrated in degree 0. We see something similar for $\mathcal{O}_X^*$. We then obtain a long exact sequence

$$\begin{array}{ccccccc}
 0 & \longrightarrow & \mathbb{Z} & \longrightarrow & \mathbb{C} & \longrightarrow & \mathbb{C}^* \\
 & & & & \searrow & & \\
 & & 0 & \longrightarrow & 0 & \longrightarrow & \text{Pic}(X) \\
 & & \searrow & & & & \\
 & & \mathbb{Z} & \longrightarrow & 0 & \longrightarrow & \dots
 \end{array}$$

We are trying to investigate $\text{Pic}(X)$. Exactness implies $\text{Pic}(X) \cong \mathbb{Z}$. One can construct an explicit isomorphism using Chern classes: $c_1(\mathcal{O}_{\mathbb{P}^1}(1)) = 1$, and we find that $\mathcal{O}_{\mathbb{P}^1}(1)$ generates the Picard group. This classifies holomorphic line bundles on $\mathbb{P}^1$.

Chapter 5

idk but we're done sheaf

Recall: a Kähler manifold is a complex manifold X , where we write the almost complex structure as J . This must carry a compatible metric, meaning J is an isometry: $g \circ (J \times J) = g$. Then X is Kähler if the fundamental form $\omega = g \circ (J \times 1)$ is closed. It turns out that Kähler manifolds have nice cohomology, so people have spent a lot of time looking for manifolds with very specific metrics. Kähler manifolds are the roughest type of this metric. For example,

$$\omega = \frac{i}{2} \sum_{i=1}^n dz_i \wedge d\bar{z}_i$$

is a closed fundamental form on $\mathbb{C}^n$. The flat metric on $\mathbb{C}^n$ is translation invariant. Recall that the metric is a section of $\text{Sym}^2 T^*X$. In particular, it can be pulled back. The notion of “translation invariance” here means that it pulls back to itself along translations. Complex tori are also Kähler, by passing g along the covering space or something.

Example 5.0.1. Let X be a 1-dimensional complex manifold. Then *any* metric is Kähler, since the fundamental form is a (real) 2-form, and thus necessarily is killed by the differential.

Proposition 5.0.2. *Let X be Kähler, and let $i : Y \hookrightarrow X$ be a complex submanifold of X . Then Y has the pullback metric induced by the Kähler metric g_X on X . This metric is Kähler.*

Proof. We simply have $g_Y = g_X|_Y$. In particular, $\omega_Y = i^*\omega_X$. Then

$$d\omega_Y = di^*\omega_X = i^*d\omega_X = 0.$$

■

Lecture 18: Kähler manifolds and the hodge star

Tue 14 Apr 2026 12:30

Proposition 5.0.2 will get a lot of mileage by proving that $\mathbb{P}^n$ is Kähler. We see that if X is compact, then $[\omega] \in H^2(X, \mathbb{C})$ is nonzero. This yields an obstruction to the existence of Kähler metrics on compact manifolds: if $H^2(X, \mathbb{C}) \cong 0$, then X is not Kähler.

Example 5.0.3. Let $\lambda \in (0, 1)$. Then $\mathbb{C}^n \setminus \{0\}$ has a $\mathbb{Z}$ -action $k \mapsto (x \mapsto \lambda^k x)$. We consider the quotient manifold by this action. As we saw earlier when considering Hopf manifolds, there is a diffeomorphism $X := \mathbb{C}^n / \mathbb{Z} \cong S^{2n-1} \times S^1$. Then in cohomology,

$$H^\bullet(X, \mathbb{C}) \cong H^\bullet(S^{2n-1} \times S^1, \mathbb{C}) \cong H^\bullet(S^{2n-1}, \mathbb{C}) \otimes_{\mathbb{C}} H^\bullet(S^1, \mathbb{C}).$$

This is of course the graded tensor product, meaning that the cohomology of spheres tells us that the cohomology is $\mathbb{C}$ concentrated in degrees 0, 1, $2n - 1$, and $2n$. We thus see that X cannot be Kähler for all $n \neq 1$. When $n = 1$, we obtain an elliptic curve, which is projective. We will see that this implies it is Kähler.

Let $(W, \langle -, - \rangle)$ be a $\mathbb{R}$ -inner product space. Then the inner product provides a canonical identification $W \cong W^*$ via $w \mapsto \langle w, - \rangle$. We also have induced inner products on all tensor, symmetric, and exterior powers of W , I think by multiplying. Let $\dim W = r$. There are maps

$$\Lambda^k W \otimes_{\mathbb{R}} \Lambda^{r-k} W \rightarrow \Lambda^r W$$

given by $\alpha \otimes \beta \mapsto \alpha \wedge \beta$. The map is nondegenerate since we can take the complement of an index. Let $\{e_i\}_{1 \leq i \leq r}$ be an orthonormal basis for W , yielding a volume form $\text{vol} := e_1 \wedge \cdots \wedge e_r$ for $\Lambda^r W$.

Definition 5.0.4. With notation as before, let $\star : \Lambda^k W \rightarrow \Lambda^{r-k} W$ be the map defined by, for any two $\alpha, \beta \in \Lambda^k W$,

$$\alpha \wedge \star \beta = \langle \alpha, \beta \rangle \text{vol}.$$

This map is defined as the *Hodge star*.

Example 5.0.5. Let $W = \text{span}\{e_1, e_2, e_3, e_4\}$ be orthonormal. Then $\star e_1$ should be something such that, when wedged with e_1 , yields the volume form. Thus $e_1 = e_2 \wedge e_3 \wedge e_4$. Similarly,

$$e_2 = -e_1 \wedge e_3 \wedge e_4,$$

$$e_3 = e_1 \wedge e_2 \wedge e_4,$$

$$e_4 = -e_1 \wedge e_2 \wedge e_3.$$

What about $\star(e_1 \wedge e_2)$? That has to be $e_3 \wedge e_4$. Similarly, $\star(e_2 \wedge e_4) = -e_1 \wedge e_3$. We have $\star(e_1 \wedge e_2 \wedge e_4) = e_3$. Finally, $\star 1 = \text{vol}$ and $\star \text{vol} = 1$.

Now the fundamental form of the vector space W live in $\omega \in \Lambda^2 W^*$. This yields an operator wedging with omega:

$$\wedge \omega : \Lambda^k W^* \rightarrow \Lambda^{k+2} W^*.$$

This yields an adjoint operator $\Lambda : \Lambda^{k+2} W^* \rightarrow \Lambda^k W^*$ defined by $\langle \alpha \wedge \omega, \beta \rangle = \langle \alpha, \Lambda \beta \rangle$. We will show that these two operators generate a complex $\mathfrak{sl}_2(\mathbb{C})$ -action on $\Lambda^\bullet W^*$. Recall that $\mathfrak{sl}_2(\mathbb{C})$ has generators

$$h := \sigma_z, \quad e := \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad f := \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$$

with commutation relations

$$[h, f] = 2f, \quad [h, e] = 2e, \quad [e, f] = h.$$

Consider a representation on W . Then h determines a map $W \rightarrow W$, and since we are working complex, it has an eigenvalue λ . Let V_λ span the eigenspace. Then

$$heV_\lambda - ehV_\lambda = [h, e]V_\lambda = 2eV_\lambda.$$

Thus

$$\lambda eV_\lambda + 2eV_\lambda = heV_\lambda = (\lambda + 2)V_\lambda.$$

Thus eV_λ is an eigenvector of h with eigenvalue $\lambda+2$, and the same proof shows fV_λ is an eigenvector with eigenvalue $\lambda-2$. We can continue this for powers by continuously raising and lowering the eigenvalue.

If this representation is a finite-dimensional irreducible representation, then at some point $e^n V_\lambda = (\lambda+n)V_\lambda$ must go to 0 (or f^n), meaning $\lambda \in \mathbb{Z}$. Then the irreducible representation looks as follows. There are 1-dimensional spaces spanned by eigenvalues of h :

$$V_n, V_{n-2}, \dots, V_{-n}.$$

Each space is 1-dimensional. The e and f operators act as ladder operators. Thus the dimensionality of the representation is the number of such V 's.

Lecture 19: Hodge theory

Thu 16 Apr 2026 12:20

Proposition 5.0.6. 1. Let $\{e_i\}_{1 \leq i \leq n}$ be an orthonormal basis for a finite dimensional inner product space $(V, \langle -, - \rangle)$. Let $\text{vol} = e_1 \wedge \dots \wedge e_n$. If $I = \{i_1 < \dots < i_k\}$ and $J = \{j_1 < \dots < j_{n-k}\}$ are indicies such that $I \amalg J = \{1, \dots, n\}$, then

$$\star e_I = \star(e_{i_1} \wedge \dots \wedge e_{i_k}) = \sigma_{I,J} \cdot e_J,$$

where $\sigma_{I,J}$ is the (sign of the) permutation $\{1, \dots, n\} \rightarrow \{i_1, \dots, i_k, j_1, \dots, j_{n-k}\}$.

2. Let $\alpha \in \Lambda^k V$ and $\beta \in \Lambda^{n-k} V$. Then

$$\langle \alpha, \star \beta \rangle = (-1)^{k(n-k)} \langle \star \alpha, \beta \rangle.$$

3. If $\alpha \in \Lambda^k V$, then $\star \star \alpha = (-1)^{k(n-k)} \alpha$.

4. The map $\star$ is an isometry of $(\Lambda^\bullet V, \langle -, - \rangle)$.

Proof. These are direct computations. ■

Suppose we have an almost complex inner product space $(V, J, \langle -, - \rangle)$. We are interested in forms $\Lambda^\bullet V^*$. We get a Hodge star $\star : \Lambda^\bullet V^* \rightarrow \Lambda^\bullet V^*$. The inner product extends to a Hermitian form $\langle -, - \rangle_h$ on $V_\mathbb{C}^*$. We noted that $(V^*)^{1,0}$ and $(V^*)^{0,1}$ are orthogonal with respect to $\langle -, - \rangle_h$. Star then defines

$$\star : \Lambda^k V_\mathbb{C}^* \rightarrow V^{n-k} V_\mathbb{C}^*$$

by

$$\alpha \wedge \star \beta = \langle \alpha, \bar{\beta} \rangle_h \text{vol}.$$

Note that $\langle \alpha, \bar{\beta} \rangle$ is nonzero if and only if α and $\bar{\beta}$ are of the same (p, q) -type, meaning if α is type (p, q) , then β must be type (q, p) . Thus

$$\star : \Lambda^{p,q} V_\mathbb{C}^* \rightarrow \Lambda^{n-q, n-p} V_\mathbb{C}^*.$$

For example, consider

$$V_\mathbb{C}^* = \text{span} \{z_1, z_2, z_3, \bar{z}_1, \bar{z}_2, \bar{z}_3\}.$$

We will compute $\star(z_1 \wedge z_2 \wedge \bar{z}_3)$. This is of type $(2, 1)$, so the Hodge star of it lives in $\Lambda^{3-1, 3-2} V_\mathbb{C}^* = \Lambda^{2,1} V_\mathbb{C}^*$. The volume form is $z_1 \wedge z_2 \wedge z_3 \wedge \bar{z}_1 \wedge \bar{z}_2 \wedge \bar{z}_3$. The naïve approach to the Hodge star would

be the compliment $\pm z_3 \wedge \bar{z}_1 \wedge \bar{z}_2$. However, this has the wrong type, and it turns out the actual thing is its complex conjugate $\pm z_1 \wedge z_2 \wedge \bar{z}_3$.

Now consider the Lefschetz operator on V . Although V is still almost complex, we now go back to working fully with its real vector space structure, and consider complexification later. Let $\omega \in \Lambda^2 V^*$ be the fundamental form. Then $L = \omega \wedge -$ is a map $\Lambda^k V^* \rightarrow \Lambda^{k+2} V^*$ (these are the real exterior powers of the dual). The adjoint of L with respect to the *real* inner product, denoted Λ , is a map $\Lambda^{k+2} V^* \rightarrow \Lambda^k V^*$. It is determined by $\langle \Lambda \alpha, \beta \rangle = \langle \alpha, L \beta \rangle$.

Since $\star$ is invertible, it makes sense to discuss $\star^{-1}$.

Proposition 5.0.7.

$$\Lambda = \star^{-1} \circ L \circ \star.$$

Proof. Let α, β be real forms. Then

$$\langle \Lambda \alpha, \beta \rangle = \langle \alpha, L \beta \rangle.$$

It's sufficient to show $\langle \alpha, L \beta \rangle \text{ vol} = \langle \beta, \star^{-1} L \star \alpha \rangle$ by uniqueness of adjoints.

We have

$$\langle \alpha, \star^{-1} L \star \beta \rangle \text{ vol} = \alpha \wedge \star (\star^{-1} L \star \beta) = \alpha \wedge L \star \beta = \alpha \wedge \omega \wedge \star \beta = \langle \beta, L \alpha \rangle \text{ vol}.$$

■

Now we complexify. These maps extend to

$$L : \Lambda^\bullet V_{\mathbb{C}}^* \rightarrow \Lambda^{\bullet+2} V_{\mathbb{C}}^*$$

$$\Lambda : \Lambda^{\bullet+2} V_{\mathbb{C}}^* \rightarrow \Lambda^\bullet V_{\mathbb{C}}^*.$$

We refine how we think of these Lefschetz operators by considering the bidegree. Recall ω has type $(1, 1)$. Thus

$$L : \Lambda^{p,q} V_{\mathbb{C}}^* \rightarrow \Lambda^{p+1,q+1} V_{\mathbb{C}}^*.$$

This implies

$$\Lambda : \Lambda^{p+1,q+1} V_{\mathbb{C}}^* \rightarrow \Lambda^{p,q} V_{\mathbb{C}}^*.$$

Our goal is to show that the cohomology of a Kähler manifold is highly constrained, and this will arise by considering an $\mathfrak{sl}_2(\mathbb{C})$ action on $\Lambda^\bullet V_{\mathbb{C}}^*$. Let $H : \Lambda^\bullet V_{\mathbb{C}}^* \rightarrow \Lambda^\bullet V_{\mathbb{C}}^*$ be defined by

$$H = \sum_{k=1}^{2n} (k - n) \Pi_k.$$

Here recall that $\Pi_k : \Lambda^\bullet V_{\mathbb{C}}^* \rightarrow \Lambda^k V_{\mathbb{C}}^*$, and that V is a $2n$ -dimensional vector space. Hence all this does is multiply a k -form by $n - k$.

Proposition 5.0.8. *The collection $\{H, L, \Lambda\}$ satisfies the $\mathfrak{sl}_2(\mathbb{C})$ commutation relations*

$$[H, L] = 2L, \quad [H, \Lambda] = -2\Lambda, \quad [L, \Lambda] = H.$$

Hence $\Lambda^\bullet V_{\mathbb{C}}^*$ is an $\mathfrak{sl}_2(\mathbb{C})$ -module.

This is proven by direct computation.

Definition 5.0.9. Let U be a finite dimensional $\mathfrak{sl}_2(\mathbb{C})$ -representation. Define $P := \{u \in U \mid fu = 0\} \subseteq U$. This is called the subspace of *primitive* vectors or *lowest weight* vectors. Here, recall that an $\mathfrak{sl}_2(\mathbb{C})$ -representation is generated by $\{e, f, h\}$.

Thus $U \cong \bigoplus_{i \geq 0} e^i P$. Note also that for any weight $\lambda \in \mathbb{Z}$ in U , we obtain $e^\lambda : U_{-\lambda} \cong U_\lambda$.

Theorem 5.0.10. If $(V, \langle -, - \rangle, J)$ is an almost complex inner product space, then

1. We can write

$$\Lambda^k V^* \cong \bigoplus_{i \geq 0} L^i P^{k-2i}.$$

I.e., any $\alpha \in \Lambda^k V^*$ can be written uniquely as

$$\alpha = \alpha_k + L\alpha_{k+2} + L^2\alpha_{k+4} + \dots, \quad \text{where } \alpha_j \in P^j.$$

2. $L^{n-k} : \Lambda^k V^* \cong \Lambda^{2n-k} V^*$.

3. If $k > n$ then $P^k = 0$. If $k \leq n$, then

$$P^k \cong \left\{ \alpha \in \Lambda^k V^* \mid L^{n-k+1} \alpha = 0 \right\}.$$

Lecture 20: More Hodge theory

Tue 21 Apr 2026 12:25

Professor Fried taught class this week.

The most important example of a Kähler manifold is $\mathbb{CP}^n$ with the Fubini-Study metric. Consider the isometry group $SU(n+1)$. Intuitively, we want to find the distance between two lines in $\mathbb{C}^{n+1} \setminus \{0\}$. This isn't quite right. Equivalently, (compactifyingish – we mod out by $z \sim \lambda z$, $|\lambda| = 1$ to recover $\mathbb{CP}^n$) we want to find the distance between circles in S^{2n+1} . We can define the distance between two sets, but this usually isn't good enough to define a metric since we don't usually get the triangle inequality. But in *this case*, it is, pending some refinements. This gives the most symmetric possible metric on $\mathbb{CP}^n$. Now we rigorize our intuition.

Start with the standard Hermitian metric $h(v, w) = \langle v, w \rangle = \sum_i v_i \bar{w}_i$ on $\mathbb{C}^{n+1}$. Define H to be the metric defined at each point x by $H_x = \frac{1}{h(x, x)} h_x$. Apparently $\mathbb{C}^*$ acts on $(\mathbb{C}^{n+1} \setminus 0, H)$ isometrically. Let $\pi : \mathbb{C}^{n+1} \setminus 0 \rightarrow \mathbb{CP}^n$ be the projection, and note that

$$\pi^* T\mathbb{CP}^n \cong T(\mathbb{C}^{n+1} \setminus 0)/L,$$

where $L \subseteq T(\mathbb{C}^{n+1} \setminus 0)$ is vectors tangent to $\mathbb{C}^*$ -orbits. I don't know what's happening so here's the metric. Let (U_i, φ_i) be a chart. Write z and w for coords on $\mathbb{CP}^n$ and $\mathbb{C}^n$, respectively. Then the Kähler form ω is defined locally by

$$\omega|_{U_i} := \frac{i}{2\pi} \partial \bar{\partial} \ln \left(\sum_{\ell=0}^n \left| \frac{z_\ell}{z_i} \right|^2 \right) \in \mathcal{A}^{1,1}(U_i)$$

$$\omega|_{U_i} := \frac{i}{2\pi} \partial \bar{\partial} \ln \left(\sum_{k=1}^n |w_k|^2 + 1 \right).$$

Note that $\omega = -\text{Im } H$ for H the metric. This is

$$\frac{i}{2} \left(\frac{\sum_{\alpha=1}^n dz_\alpha \wedge d\bar{z}_\alpha}{1 + \sum_{\alpha=1}^n |z_\alpha|^2} - \frac{\sum_1^n \sum_\alpha \bar{z}_\alpha dz_\alpha \wedge \sum_1^n z_\alpha d\bar{z}_\alpha}{(?)^2} \right).$$

Idk what that is. The book uses

$$h_{ij} \left(1 + \sum_i |w_i|^2 \right) \delta_{ij} - \bar{w}_i w_j.$$

And

$$\omega = \frac{1}{\left(1 + \sum_i |w_i|^2 \right)^2} \sum_{i,j} h_{ij} dw_i \wedge d\bar{w}_j.$$

See page 118, there's more.

Now the Kähler identities. We have operators $d, \partial, \bar{\partial}$, and they have hermitian adjoints. These are important 1st order operators. We have important 0th order operators $L, \Lambda, \star$. We have important 2nd order operators

$$\Delta = dd^* + d^*d, \quad \Delta_{\partial} = \partial\partial^* + \partial^*\partial, \quad \Delta_{\bar{\partial}} = \bar{\partial}\bar{\partial}^* + \bar{\partial}^*\bar{\partial}.$$

Ok but wtf is the adjoint of a differential operator? Well it's just the adjoint under the L^2 inner product. It can be derived via the Leibniz identity.

Lecture 21: Hodge decomposition

Thu 23 Apr 2026 12:30

Define the inner product on $\mathcal{A}_X^k(U)$ by

$$\langle \alpha, \beta \rangle_U := \int_U \alpha \wedge \star \bar{\beta}.$$

We claim that

$$\langle \partial\alpha, \beta \rangle_X = \langle \alpha, -\star \partial \star \beta \rangle =: \langle \alpha, \partial^* \beta \rangle.$$

To show this, let α be a (p, q) -form, and note that $\partial\alpha$ is a $(p+1, q)$ -form. Thus we have $\star\beta \in \mathcal{A}_X^{n-p-1, n-q}$. This implies $\beta \in \mathcal{A}_X^{q, p+1}$. Then

$$\langle \partial\alpha, \beta \rangle = \int_X \partial\alpha \wedge \star \bar{\beta}.$$

Stokes shows that

$$0 = \int_X \partial(\alpha \wedge \star \bar{\beta}) = \int_X \partial\alpha \wedge \star \bar{\beta} + (-1)^{|\alpha|} \int_X \alpha \wedge \partial \star \bar{\beta}$$

since $d = \partial + \bar{\partial}$ and $\bar{\partial} = 0$ apparently. Anyways some math shows

$$\partial^* = -\star \partial \star$$

$$\bar{\partial}^* = -\star \bar{\partial} \star$$

$$d^* = -\star d \star$$

We now have the Kähler identities.

1. $[L, \partial] = 0$
2. $[L, \bar{\partial}] = 0$

3. $[\Lambda, \partial^*] = 0$
4. $[\Lambda, \bar{\partial}^*] = 0$
5. $[L, \partial^*] = i\bar{\partial}$
6. $[L, \bar{\partial}^*] = -i\partial$
7. $[\Lambda, \partial] = i\bar{\partial}^*$
8. $[\Lambda, \bar{\partial}] = -i\partial^*$
9. $\partial\bar{\partial}^* + \bar{\partial}^*\partial = 0$
10. $\Delta_d = (\partial + \bar{\partial})(\partial^* + \bar{\partial}^*) + (\partial^* + \bar{\partial}^*)(\partial + \bar{\partial})$
11. $\Delta_d = \Delta_\partial + \Delta_{\bar{\partial}}$
12. $\Delta_d d = d\Delta_d$

Now we want to look for global topological data on compact Kähler manifolds. All holomorphic p -forms, aka $(p, 0)$ -forms, are closed on Kähler manifolds. It is even true that the cohomology class of any nonzero holomorphic p -form is nonzero. What this means is that there are not a lot of holomorphic forms on compact Kähler manifolds.

Lecture 22: Hodge theory and stuff

Tue 28 Apr 2026 12:30

Theorem 5.0.11. *If (X, g) is Kähler, then*

$$\Delta_\partial = \Delta_{\bar{\partial}} = \frac{1}{2}\Delta_d.$$

Proposition 5.0.12. *Let (X, g) Kähler. Then α is d-harmonic iff $\|\alpha\|$ is minimal in its cohomology class.*

Theorem 5.0.13 (Hodge). *Let (X, g) a compact complex manifold with a compatible metric. Then the 3 harmonic spaces of harmonic forms $\mathcal{H}_d \subseteq \mathcal{A}^k(X)$, $\mathcal{H}_{\bar{\partial}}^{p,q} \subseteq \mathcal{A}^{p,q}(X)$, and $\mathcal{H}_{\partial}^{p,q} \subseteq \mathcal{A}^{p,q}(X)$ are finite dimensional. Moreover, there are orthogonality relations*

$$\begin{aligned}\mathcal{A}^k(X) &= d\mathcal{A}^{k-1}(X) \oplus \mathcal{H}_d^k(X) \oplus d^*\mathcal{A}^{k+1}(X) \\ \mathcal{A}^{p,q}(X) &= \partial\mathcal{A}^{p-1,q}(X) \oplus \mathcal{H}_{\bar{\partial}}^{p,q} \oplus \partial^*\mathcal{A}^{p+1,q}(X) \\ \mathcal{A}^{p,q}(X) &= \bar{\partial}\mathcal{A}^{p,q-1}(X) \oplus \mathcal{H}_{\partial}^{p,q} \oplus \bar{\partial}^*\mathcal{A}^{p,q+1}(X).\end{aligned}$$

Using Hodge's theorem, we compute

Lemma 5.0.14.

$$\text{Ker} \left(d : \mathcal{A}^k(X) \rightarrow \mathcal{A}^{k+1}(X) \right) = \mathcal{H}_d^k(X) \oplus d\mathcal{A}^{k-1}.$$

The analogous statements hold for $\partial, \bar{\partial}$.

Corollary 5.0.15. *If (X, g) is a compact Hermitian manifold, then there are canonical isomorphisms*

$$\begin{aligned}\mathcal{H}_d^k(X, g) &\cong H_{dR}^k(X) \cong H_{\text{Sing}}^k(X, \mathbb{R}) \\ \mathcal{H}_{\bar{\partial}}^{p,q}(X, g) &\cong H_{\bar{\partial}}^{p,q}(X) \cong H_{\text{shf}}^q(X, \Omega_{\text{hol}}^p) \\ \mathcal{H}_{\partial}^{p,q}(X, g) &\cong H_{\partial}^{p,q}(X) \cong H_{\text{shf}}^p(X, \Omega_{\text{antihol}}^p).\end{aligned}$$

In particular, the Hodge theorem implies that each cohomology class on a Kähler manifold has a canonical harmonic representative.

Theorem 5.0.16. *Let (X, g) be a compact Kähler manifold. Then there is a decomposition of de Rham = singular cohomology*

$$H^k(X, \mathbb{C}) \cong \bigoplus_{p+q=k} H^{p,q}(X).$$

This is independent of the metric. Additionally,

$$H^{q,p}(X) = \overline{H^{p,q}(X)}.$$

$$H^{p,q}(X) \cong H^{n-p, n-q}(X).$$

Proof. Fix a Kähler metric g on X . We get isomorphisms $\mathcal{H}_{\bar{\partial}(X)}^{p,q} \cong H_{\bar{\partial}}^{p,q}(X)$, $\mathcal{H}_{\partial}^{p,q}(X) \cong H_{\partial}^{p,q}(X)$, and $\mathcal{H}_d^k(X) \cong H^k(X)$. Take $\Delta_d = 2\Delta_{\bar{\partial}} = 2\Delta_{\partial}$. Take $\alpha \in \mathcal{H}_d^k(X)$. Then

$$\alpha = \sum_{p+q=k} \alpha^{p,q}, \quad \alpha^{p,q} \in \mathcal{A}^{p,q}(X).$$

Then

$$0 = \Delta_d \alpha = \sum_{p+q=k} \Delta_d \alpha^{p,q} \implies \Delta_d \alpha^{p,q} = 0 \implies \Delta_{\bar{\partial}} \alpha^{p,q} = 0.$$

Thus the first statement follows. Now we show that this isomorphism is independent of g . ■

This symmetry allows us to organize the cohomology into nice diagrams. Let $h^{p,q} := \dim_{\mathbb{C}} H^{p,q}(X)$. Let $\dim_{\mathbb{C}} X = n$. One constructs a diamond

$$\begin{array}{ccccc} & & h^{n,n} & & \\ & \swarrow & | & \searrow & \\ h^{n/2,0} & & & & h^{0,n/2} \\ & \swarrow & | & \searrow & \\ & & h^{0,0} & & \end{array}$$

This diamond is invariant under horizontal and vertical flips, and thus under 180 degree rotations. This is called the Hodge diamond.

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We will now compute $\text{Pic}(\mathbb{P}^n)$. Recall that this is the group of isomorphism classes of line bundles, and for $\mathbb{P}^1$ it was $\mathbb{Z}$. Let $X = \mathbb{P}^n$. Then Pic is given by the sheaf cohomology $\text{Pic}(X) = H^1(X, \mathcal{O}_X^*)$. We use the short exact sequence

$$0 \longrightarrow \mathbb{Z} \longrightarrow \mathcal{O}_X \xrightarrow{\exp} \mathcal{O}_X^* \longrightarrow 0$$

which yields the long exact sequence

$$\cdots \longrightarrow H^n(X, \mathbb{Z}) \longrightarrow H^n(X, \mathcal{O}_X) \longrightarrow H^n(X, \mathcal{O}_X^*) \longrightarrow H^{n+1}(X, \mathbb{Z}) \longrightarrow \cdots$$

We use $H^q(X, \mathcal{O}_X) = H^{0,q}(X, \mathcal{O}_X)$. In small degrees, this reduces to

$$0 \longrightarrow \text{Pic}(X) \longrightarrow \mathbb{Z} \longrightarrow 0.$$

This proves $\text{Pic}(X) \cong \mathbb{Z}$.

Lemma 5.0.17 ($\partial\bar{\partial}$). *Let (X, g) a compact Kähler manifold, or a blow up of a Kähler manifold. Let $\alpha \in \mathcal{A}^{p,q}$ be d-closed. Then the following are equivalent.*

1. α is d-exact.
2. α is ∂ -exact.
3. α is $\bar{\partial}$ -exact.
4. α is $\partial\bar{\partial}$ -exact.
5. α is orthogonal to $\mathcal{H}^{p,q}(X, g)$ for an arbitrary Kähler metric.

Proof. $(1, 2, 3, 4 \implies 5)$ Since α is pure of type (p, q) , it is d-closed iff it is ∂ and $\bar{\partial}$ -closed. This implies condition 5 by Hodge's theorem. It's enough to show $(5 \implies 4)$. This also follows by Hodge's theorem. ■

Since L, Λ commute with any of these Laplacians on a compact Kähler manifold, they will descend to the harmonic space, which we have seen is the same as cohomology. So for example,

$$L : H^{p,q}(X) \rightarrow H^{p+1,q+1}(X), \quad \Lambda : H^{p+1,q+1}(X) \rightarrow H^{p,q}.$$

Note that they depend only on the cohomology class of the fundamental form.

Theorem 5.0.18 (Hard Lefschetz). *The map*

$$L^{n-k} : H^k(X) \rightarrow H^{2n-k}(X)$$

is an isomorphism.

This is because L is an $\mathfrak{sl}_2$ -representation and gives an isomorphism with weight spaces. We already knew this isomorphism existed by Poincare duality, but the interesting bit is that it is given by wedging with $[\omega]^{n-k}$.

This gives us a Lefschetz decomposition. Define $P^k(X, \mathbb{C})$ to be the kernel of $\Lambda : H^k(X, \mathbb{C}) \rightarrow H^{k+2}(X, \mathbb{C})$. We can of course take a (p, q) -refinement of this.

Theorem 5.0.19 (Lefschetz decomposition). *We have*

$$H^k(X, \mathbb{C}) \cong \bigoplus_{i \geq 0} L^i P^{k-2i}(X, \mathbb{C}).$$

Similarly for Dolbeault cohomology,

$$H^{p,q}(X) \cong \bigoplus_{i \geq 0} L^i P^{p-i, q-i}(X).$$

Theorem 5.0.20 (Lefschetz theorem on $(1, 1)$ -classes). *Let $L \rightarrow X$ be a complex line bundle (smooth line bundle with fibers isomorphic to $\mathbb{C}$ – not necessarily holomorphic). Then L is completely classified by $c_1(L) \in H^2(X, \mathbb{Z})$.*

Proof. Recall that complex line bundles L on X are the homotopy classes of maps $X \rightarrow BL$ to the classifying space. In this case, it is $BL = \mathbb{C}\mathbb{P}^\infty$. But $\mathbb{C}\mathbb{P}^\infty$ is an Eilenberg-MacLane space $K(\mathbb{Z}, 2)$, which classifies cohomology by homotopy classes of maps into it. Thus $[X, \mathbb{C}\mathbb{P}^\infty] \cong H^2(X, \mathbb{Z})$. Let $H^{1,1}(X, \mathbb{Z})$ be the classes of $H^2(X, \mathbb{Z})$ such that they map to $H^{1,1}(X) \subseteq H^2(X, \mathbb{C})$. The statement is that these are exactly the first Chern classes, meaning it suffices to show that the image of $c_1 : \text{Pic}(X) \rightarrow H^2(X, \mathbb{Z})$ is $H^{1,1}(X, \mathbb{Z})$. ■

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